

Complex Geometry

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Chapter 1 Foundations of Kähler Geometry

1 Complex Manifolds and Complex Structures

In this section, we mainly introduce some basic concepts in complex geometry.

Definition 1.1.1. A function $F : U \subset \mathbb{C}^n \rightarrow \mathbb{C}^m$ is called **holomorphic** iff the differential dF of F as a real map $F : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2m}$ satisfies $j_m \circ dF = dF \circ j_n$, where j_m, j_n are endomorphisms of $\mathbb{R}^{2m}, \mathbb{R}^{2n}$ respectively:

$$\begin{pmatrix} 0 & -I_m \\ I_m & 0 \end{pmatrix}; \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$$

Definition 1.1.2. A **complex manifold** of complex dimension m is a topological space M with an open covering $\mathcal{U}$ such that for any point $x \in M$, there exists $U \subset \mathcal{U}$ containing x and a homeomorphism $\phi_U : U \rightarrow \tilde{U} \subset \mathbb{C}^m$, such that for every intersecting $U \cap V$, the map between open subsets of $\mathbb{C}^m$:

$$\phi_{UV} = \phi_U \circ \phi_V^{-1} : \phi_V(U \cap V) \rightarrow \phi_U(U \cap V)$$

is holomorphic. A pair $\{U, \phi_U\}$ is called a **chart** and the collections of all charts is called **holomorphic structure**.

Then we say a function $F : M \rightarrow \mathbb{C}$ is holomorphic if $F \circ \phi_U^{-1}$ is holomorphic for any $U \in \mathcal{U}$.

Definition 1.1.3. A $(1,1)$ -tensor J on a differential manifold M satisfying $J^2 = -id$ is called an **almost complex structure**, and the pair (M, J) is then referred to as

almost complex manifold. Moreover, if the almost complex structure J comes from a holomorphic structure, then we call J a complex structure.

Actually, we can construct the complex structure on a complex manifold directly: for every $X \in T_x M$, we choose $U \in \mathcal{U}$ containing x and define

$$J_U(X) = (d\phi_U)^{-1} \circ j \circ (d\phi_U)(X) \quad (1.1)$$

Then

$$\begin{aligned} J_V(X) &= (d\phi_V)^{-1} \circ j \circ (d\phi_V)(X) \\ &= (d\phi_V)^{-1} \circ j \circ (d\phi_{VU}) \circ (d\phi_U)(X) \\ &= (d\phi_V)^{-1} \circ (d\phi_{VU}) \circ j \circ (d\phi_U)(X) \\ &= (d\phi_U)^{-1} \circ j \circ (d\phi_U)(X) \\ &= J_U(X) \end{aligned}$$

Which means that the collection of $\{J_U\}$ is a well defined tensor on M , we denote it by J , and J satisfies $J^2 = -id$.

Now we introduce the complexified tangent bundle. Let (M, J) be an almost complex manifold, define

$$TM^{\mathbb{C}} = TM \otimes_{\mathbb{R}} \mathbb{C}$$

We extend real endomorphisms and differential operators from TM to $TM^{\mathbb{C}}$ by $\mathbb{C}$ -linearity. Denote the eigenbundle of $TM^{\mathbb{C}}$ corresponding to the eigenvalue $\sqrt{-1}$ (resp. $-\sqrt{-1}$) by $T^{1,0}M$ (resp. $T^{0,1}M$), then we have

$$\begin{aligned} T^{1,0}M &= \{X - \sqrt{-1}JX : X \in TM\} \\ T^{0,1}M &= \{X + \sqrt{-1}JX : X \in TM\} \end{aligned}$$

and

$$TM^{\mathbb{C}} = T^{1,0}M \oplus T^{0,1}M$$

Suppose an almost complex structure J comes from a holomorphic structure on complex manifold M , consider a local chart $\{U, \phi_U\}$ and let $z_\alpha = x_\alpha + \sqrt{-1}y_\alpha$ be the α -component of ϕ_U . Denote the standard basis of $\mathbb{R}^{2m}$ by $\{e_1, e_2, \dots, e_{2m}\}$, then by definition:

$$\frac{\partial}{\partial x_\alpha} = (d\phi_U)^{-1}(e_\alpha); \quad \frac{\partial}{\partial y_\alpha} = (d\phi_U)^{-1}(e_{m+\alpha})$$

Holds. Moreover, $j(e_\alpha) = e_{m+\alpha}$, combine with (1.10) we have

$$J\left(\frac{\partial}{\partial x_\alpha}\right) = \frac{\partial}{\partial y_\alpha} \quad (1.2)$$

$$J\left(\frac{\partial}{\partial y_\alpha}\right) = -\frac{\partial}{\partial x_\alpha} \quad (1.3)$$

Make the following notations

$$\frac{\partial}{\partial z_\alpha} = \frac{1}{2} \left(\frac{\partial}{\partial x_\alpha} - \sqrt{-1} \frac{\partial}{\partial y_\alpha} \right); \quad \frac{\partial}{\partial \bar{z}_\alpha} = \frac{1}{2} \left(\frac{\partial}{\partial x_\alpha} + \sqrt{-1} \frac{\partial}{\partial y_\alpha} \right)$$

From (1.11), (1.12) we obtain that $\left\{ \frac{\partial}{\partial z_\alpha} \right\}$ and $\left\{ \frac{\partial}{\partial \bar{z}_\alpha} \right\}$ are local sections of $T^{1,0}M$ and $T^{0,1}M$ respectively. Actually, they form a local basis around x . Now let Z, W be two local sections of $T^{0,1}M$ and write $Z = Z_\alpha \frac{\partial}{\partial \bar{z}_\alpha}$, $W = W_\alpha \frac{\partial}{\partial \bar{z}_\alpha}$, we compute the Lie bracket:

$$[Z, W] = \left[Z_\alpha \frac{\partial}{\partial \bar{z}_\alpha}, W_\beta \frac{\partial}{\partial \bar{z}_\beta} \right] = Z_\alpha \frac{\partial W_\beta}{\partial \bar{z}_\alpha} \frac{\partial}{\partial \bar{z}_\beta} - W_\alpha \frac{\partial Z_\beta}{\partial \bar{z}_\alpha} \frac{\partial}{\partial \bar{z}_\beta}$$

Which is a local section of $T^{0,1}M$. The above computation implies that $T^{0,1}M$ is involutive, by Frobenius Theorem, it's integrable.

The converse is true either, i.e. the distribution $T^{0,1}M$ is integrable implies the almost complex structure J comes from a holomorphic structure, we omit the proof. In conclusion, we have deduced the famous Newlander-Nirenberg theorem:

Theorem 1.1.4. *Let (M, J) be an almost complex manifold, then the almost complex structure J comes from a holomorphic structure iff the distribution $T^{0,1}M$ is integrable.*

Now we introduce the complexified exterior bundle likewise. Let (M, J) be an almost complex manifold, define

$$\Lambda_{\mathbb{C}}^* M = \Lambda^* M \otimes_{\mathbb{R}} \mathbb{C}$$

And two subbundles of $\Lambda_{\mathbb{C}}^1 M$:

$$\Lambda^{1,0} M = \left\{ \xi \in \Lambda_{\mathbb{C}}^1 M : \xi(Z) = 0, \forall Z \in T^{0,1} M \right\}$$

$$\Lambda^{0,1} M = \left\{ \xi \in \Lambda_{\mathbb{C}}^1 M : \xi(Z) = 0, \forall Z \in T^{1,0} M \right\}$$

The sections of these subbundles are called forms of type $(1, 0)$ or type of $(0, 1)$ respectively. For complexified exterior bundle, we still have the decomposition:

$$\Lambda_{\mathbb{C}}^1 M = \Lambda^{1,0} M \oplus \Lambda^{0,1} M$$

Where

$$\begin{aligned}\Lambda^{1,0}M &= \{\omega - \sqrt{-1}\omega \circ J : \omega \in \Lambda^1M\} \\ \Lambda^{0,1}M &= \{\omega + \sqrt{-1}\omega \circ J : \omega \in \Lambda^1M\}\end{aligned}$$

We need more notations. Let us denote the k -th exterior power of $\Lambda^{1,0}$ (resp. $\Lambda^{0,1}$) by $\Lambda^{k,0}$ (resp. $\Lambda^{0,k}$) and let $\Lambda^{p,q}$ denote the tensor product $\Lambda^{p,0} \otimes \Lambda^{0,q}$. The exterior power of a direct sum of vector spaces can be described as $\Lambda^k(E \oplus F) \cong \bigoplus_{i=0}^k \Lambda^i E \wedge \Lambda^{k-i} F$, thus we have

$$\begin{aligned}\Lambda_{\mathbb{C}}^k M &\cong \bigoplus_{i=0}^k \Lambda^i(\Lambda^{1,0}M) \wedge \Lambda^{k-i}(\Lambda^{0,1}M) \\ &= \bigoplus_{i=0}^k \Lambda^{i,0}M \wedge \Lambda^{0,k-i}M \\ &= \bigoplus_{p+q=k} \Lambda^{p,q}M\end{aligned}$$

Where sections of $\Lambda^{p,q}M$ are called forms of type (p, q) . Now we claim two facts without proof:

- A complex value k -form ω is a section of $\Lambda^{k,0}M$ iff $i_Z\omega$ vanishes for any $Z \in T^{0,1}M$.
- A complex value k -form ω is a section of $\Lambda^{p,q}M$ iff it vanishes whenever applied to $p+1$ vectors from $T^{1,0}M$ and $q+1$ vectors from $T^{0,1}M$.

If J is a complex, we can describe above spaces in terms of local coordinates. We consider in a local chart $\{U, \phi_U\}$, and let $z_\alpha = x_\alpha + \sqrt{-1}y_\alpha$ be the α -component of ϕ_U . If we extend exterior differential operator d on functions by $\mathbb{C}$ -linearity, we get complex value forms $dz_\alpha = dx_\alpha + \sqrt{-1}dy_\alpha$ and $d\bar{z}_\alpha = dx_\alpha - \sqrt{-1}dy_\alpha$. Similarly $\{d\bar{z}_\alpha\}$ is a local basis of $\Lambda^{0,1}M$ and $\{dz_\alpha\}$ is a local basis of $\Lambda^{1,0}M$. Meanwhile, a local basis for $\Lambda^{p,q}M$ can be written as

$$\{dz_{i_1} \wedge dz_{i_2} \wedge \cdots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge d\bar{z}_{j_2} \wedge \cdots \wedge d\bar{z}_{j_q} : i_1 < \cdots < i_p; j_1 < \cdots < j_q\}$$

To every almost complex structure J one can associate a $(2, 1)$ -tensor N^J called the Nijenhuis tensor, which defined as

$$N^J(X, Y) = [X, Y] + J[JX, Y] + J[X, JY] - [JX, JY]$$

Where $X, Y \in \Gamma(TM)$. We state a pivotal proposition which relates Nijenhuis tensor, exterior differential operator and almost complex structure:

Proposition 1.1.5. *Let J be an almost complex structure on M^{2m} , then the following statements are equivalent:*

- 1) J is a complex structure.
- 2) The distribution $T^{0,1}M$ is integrable.
- 3) $d\Gamma(\Lambda^{1,0}M) \subset \Gamma(\Lambda^{2,0}M \oplus \Lambda^{1,1}M)$.
- 4) $d\Gamma(\Lambda^{p,q}M) \subset \Gamma(\Lambda^{p+1,q}M \oplus \Lambda^{p,q+1}M)$ for any $0 \leq p, q \leq m$.
- 5) $N^J = 0$.

Proof. The equivalence between 1) and 2) is just Newlander-Nirenberg theorem.

First we prove 2) is equivalent to 3). Let ω be a section of $\Lambda^{1,0}M$. We will use a fact: The $\Lambda^{0,2}M$ -component of $d\omega$ vanishes iff $d\omega$ vanishes on $T^{0,1}M \times T^{0,1}M$. Take $Z, W \in T^{0,1}M$ and extend to a local section of $T^{0,1}M$, we compute

$$d\omega(Z, W) = Z(\omega(W)) - W(\omega(Z)) - \omega([Z, W]) = -\omega([Z, W])$$

Then

$$\begin{aligned} [Z, W] &\in T^{0,1}M \quad \forall Z, W \in T^{0,1}M \\ \Leftrightarrow \omega([Z, W]) &= 0 \quad \forall Z, W \in T^{0,1}M, \omega \in \Lambda^{1,0}M \\ \Leftrightarrow d\omega(Z, W) &= 0 \quad \forall Z, W \in T^{0,1}M, \omega \in \Lambda^{1,0}M \end{aligned}$$

Then we prove 3) is equivalent to 4). 4) implies 3) is obviously. Conversely, suppose 3) holds. Then by conjugation,

$$d\Gamma(\Lambda^{0,1}M) \subset \Gamma(\Lambda^{1,1}M \oplus \Lambda^{0,2}M)$$

Any element τ in $\Lambda^{p,q}M$ can be locally written as sum of elements $\omega_1 \wedge \cdots \wedge \omega_p \wedge \bar{\eta}_1 \wedge \cdots \wedge \bar{\eta}_q$, then apply Leibniz' rule and assumption 3), we obtain 4).

We prove 2) is equivalent to 5) and all the statements are equivalent eventually. Let $X, Y \in \Gamma(TM)$ be local vector fields and denote Lie bracket $[X + \sqrt{-1}JX, Y + \sqrt{-1}JY]$ by Z . Since

$$\begin{aligned} Z - \sqrt{-1}JZ &= ([X, Y] + \sqrt{-1}[JX, Y] + \sqrt{-1}[X, JY] - [JX, JY]) \\ &\quad - (\sqrt{-1}J[X, Y] - J[JX, Y] - J[X, JY] - \sqrt{-1}J[JX, JY]) \\ &= N^J(X, Y) - \sqrt{-1}JN^J(X, Y) \end{aligned}$$

Then $Z \in T^{0,1}M$ is equivalent to $N^J = 0$, which implies distribution $T^{0,1}M$ is integrable is equivalent to $N^J = 0$. $\square$ $\square$

In the reminder of this section, we put (M, J) a complex manifold of dimension m . Use proposition 2.2, for every fixed (p, q) we may define two differential operators $\partial : \Gamma(\Lambda^{p,q}M) \rightarrow \Gamma(\Lambda^{p+1,q}M)$ and $\bar{\partial} : \Gamma(\Lambda^{p,q}M) \rightarrow \Gamma(\Lambda^{p,q+1}M)$ by $d = \partial + \bar{\partial}$. Since $d^2 = 0$, we deduce

$$\partial^2 = 0; \bar{\partial}^2 = 0; \partial\bar{\partial} + \bar{\partial}\partial = 0 \quad (1.4)$$

We state the following significant local $\sqrt{-1}\partial\bar{\partial}$ -Lemma:

Proposition 1.1.6. *Let $\omega \in \Lambda^{1,1}M \cap \Lambda^2M$ be a real 2-form of type $(1, 1)$ on a complex manifold. Then ω is closed iff every point $x \in M$ has an open neighbourhood U s.t. $\omega|_U = \sqrt{-1}\partial\bar{\partial}u$ for some real function u .*

Proof. ' $\Leftarrow$ ': from (2.6),

$$d\omega = d(\sqrt{-1}\partial\bar{\partial}u) = \sqrt{-1}(\partial + \bar{\partial})\partial\bar{\partial}u = \sqrt{-1}(\partial^2\bar{\partial} - \partial\bar{\partial}^2)u = 0$$

' $\Rightarrow$ ': we need $\bar{\partial}$ -Poincaré Lemma as below

Lemma 1.1.7. *A $\bar{\partial}$ -closed $(0, 1)$ -form is locally $\bar{\partial}$ -exact.*

Let ω be a closed real form of type $(1, 1)$, Poincaré Lemma implies there exists locally a 1-form η , s.t. $\omega = d\eta$. Decompose η in forms of type $(1, 0)$ and $(0, 1)$: $\eta = \eta^{1,0} + \eta^{0,1}$, we write

$$\omega = d\eta = \partial\eta^{1,0} + (\bar{\partial}\eta^{1,0} + \partial\eta^{0,1}) + \bar{\partial}\eta^{0,1}$$

Hence $\partial\eta^{1,0} = \bar{\partial}\eta^{0,1} = 0$ and $\omega = \bar{\partial}\eta^{1,0} + \partial\eta^{0,1}$. Lemma 2.1 yields a local function f s.t. $\eta^{0,1} = \bar{\partial}f$, then by conjugation $\eta^{1,0} = \partial\bar{f}$. Eventually,

$$\omega = \bar{\partial}\eta^{1,0} + \partial\eta^{0,1} = \partial\bar{\partial}f + \bar{\partial}\partial\bar{f} = \sqrt{-1}\partial\bar{\partial}(2\text{Im}f)$$

$u = 2\text{Im}f$ is all we need. $\square$ $\square$

2 Chern Connection and Chern Curvature

In the first part of this section, we introduce decompositions of connection $\nabla = \nabla^{1,0} + \nabla^{0,1}$ and exterior differential $d = \partial + \bar{\partial}$, and then derive the existence and uniqueness of Chern connection. Recall some basic concepts.

Definition 1.2.1. *Let M be a complex manifold and $\pi : E \rightarrow M$ a complex vector bundle (i.e. each fiber is a complex vector space), E is called **holomorphic vector bundle** if there exists a trivialization with holomorphic transition functions.*

Example 1.2.2. *The tangent bundle of a complex manifold M^{2m} is a holomorphic vector bundle of rank m . To see this, take a local chart $\{U, \phi_U\}$ on M and define local trivialization $\psi_U : TM|_U \rightarrow U \times \mathbb{C}^m$ as $\psi_U(X_p) = (p, d\phi_U(X_p))$, then the transition functions $g_{UV} = d\phi_U \circ (d\phi_V)^{-1}$ are holomorphic clearly.*

Example 1.2.3. *Likewise, the cotangent bundle, and more generally the exterior bundles $\Lambda^{p,0}M$ are holomorphic.*

In the previous section, we introduced $\bar{\partial} : \Gamma(\Lambda^{p,q}M) \rightarrow \Gamma(\Lambda^{p,q+1}M)$ operates on differential forms on M , consider holomorphic E , now we define $\bar{\partial}$ operates on sections of E with differential forms on M as coefficients. Denote bundle $\Lambda^{p,q}M \otimes E$ by $\Lambda^{p,q}E$, the sections of this bundle are (p,q) -forms valued in E . Take any section $\sigma \in \Gamma(\Lambda^{p,q}E)$, in a local basis frame fields $\{e_k\}$ we write locally $\sigma = \sum \omega_k e_k$, then we define $\bar{\partial}\sigma \stackrel{\text{set}}{=} \sum \bar{\partial}\omega_k e_k$. To verify the rationality of this definition, consider another local basis frame fields $\{e'_k\}$ and there exists holomorphic functions $\{a_{tk}\}$ such that $e_k = \sum a_{tk} e'_t$, then coefficient functions are interrelated via $\omega'_k = \sum \omega_t a_{tk}$. We compute

$$\sum \bar{\partial}\omega'_k e'_k = \sum \bar{\partial} \left(\sum \omega_t a_{tk} \right) e'_k = \sum \bar{\partial}\omega_t a_{tk} e'_k = \sum \bar{\partial}\omega_t e_t$$

Therefore, $\bar{\partial}\sigma$ is independent of choice of local basis. By construction, one has $\bar{\partial}^2 = 0$ and $\bar{\partial}$ satisfies Leibniz rule:

$$\bar{\partial}(\omega \wedge \sigma) = \bar{\partial}\omega \wedge \sigma + (-1)^{|\omega|} \omega \wedge \bar{\partial}\sigma \quad (1.5)$$

for any $\omega \in \Gamma(\Lambda^{p,q}M)$ and $\sigma \in \Gamma(\Lambda^{r,s}E)$.

Inversely, an operator $\bar{\partial} : \Gamma(\Lambda^{p,q}E) \rightarrow \Gamma(\Lambda^{p,q+1}E)$ on a complex vector bundle E satisfying

the Leibniz rule is called a **pseudo-holomorphic structure**, if, moreover, $\bar{\partial}^2 = 0$, then $\bar{\partial}$ is called a **holomorphic structure**. A section σ in a pseudo-holomorphic vector bundle $(E, \bar{\partial})$ is called holomorphic if $\bar{\partial}\sigma = 0$.

Actually, a vector bundle is holomorphic or not depends deeply on whether it admits a holomorphic structure, that is

Theorem 1.2.4. *A complex vector bundle E is holomorphic iff it admits a holomorphic structure $\bar{\partial}$.*

We omit the proof.

Suppose now that M is a complex manifold, and E a complex vector bundle over M . Consider two projections $\pi^{1,0} : \Lambda^1(E) \rightarrow \Lambda^{1,0}(E)$ and $\pi^{0,1} : \Lambda^1(E) \rightarrow \Lambda^{0,1}(E)$, then for any connection ∇ on E , one can consider its $(1,0)$ -component $\nabla^{1,0} = \pi^{1,0} \circ \nabla$ and $(0,1)$ -component $\nabla^{0,1} = \pi^{0,1} \circ \nabla$. From proposition 2.2, we can extend these two operators to $\nabla^{1,0} : \Gamma(\Lambda^{p,q}(E)) \rightarrow \Gamma(\Lambda^{p+1,q}(E))$ and $\nabla^{0,1} : \Gamma(\Lambda^{p,q}(E)) \rightarrow \Gamma(\Lambda^{p,q+1}(E))$ satisfying Leibniz rule:

$$\begin{aligned}\nabla^{1,0}(\omega \otimes \sigma) &= \partial\omega \otimes \sigma + (-1)^{|\omega|}\omega \wedge \nabla^{1,0}\sigma \\ \nabla^{0,1}(\omega \otimes \sigma) &= \bar{\partial}\omega \otimes \sigma + (-1)^{|\omega|}\omega \wedge \nabla^{0,1}\sigma\end{aligned}$$

Then obviously $\nabla^{0,1}$ is a pseudo-holomorphic structure on bundle E for any connection ∇ .

For the corresponding curvature R^∇ of connection $\nabla = \nabla^{1,0} + \nabla^{0,1}$, we have

$$R^\nabla = \nabla^2 = (\nabla^{1,0} + \nabla^{0,1})^2 = (\nabla^{1,0})^2 + (\nabla^{1,0}\nabla^{0,1} + \nabla^{0,1}\nabla^{1,0}) + (\nabla^{0,1})^2$$

Therefore the $\Lambda^{0,2}$ -part of curvature R^∇ is $(\nabla^{0,1})^2$. Theorem 2.5 implies that if the $\Lambda^{0,2}$ -part of curvature R^∇ vanishes, then bundle E is a holomorphic bundle with holomorphic structure $\bar{\partial} = \nabla^{0,1}$. Conversely, suppose E is a holomorphic vector bundle with holomorphic structure $\bar{\partial}$, is there possible to construct a connection ∇ s.t. $\bar{\partial} = \nabla^{0,1}$?

Above conjecture is true for Hermitian vector bundle. We introduce the definition of Hermitian structure

Definition 1.2.5. *Suppose M is a differential manifold and $E \rightarrow M$ a complex vector bundle, a **Hermitian structure** h on E is a smooth field of Hermitian products on fibers of E , that is, for every $x \in M$, $h : E_x \times E_x \rightarrow \mathbb{C}$ satisfies*

- for every $v \in E_x$, $h(u, v)$ is $\mathbb{C}$ -linear in u .
- for $u, v \in E_x$, $h(u, v) = \overline{h(v, u)}$
- for every $u \in E_x, u \neq 0$, $h(u, u) > 0$.
- for smooth sections u, v , $h(u, v)$ is a smooth function on M .

Remark 1. From above definition, clearly for every $u \in E_x$, $h(u, v)$ is $\mathbb{C}$ -anti-linear in v .

Remark 2. Actually, one may think h as to a $\mathbb{C}$ -anti-linear isomorphism $h : E \rightarrow E^*$ via $v \mapsto h(\cdot, v)$. Then we can define the metric dual connection of ∇^E on complex vector bundle E as $\nabla^{E,*} \stackrel{\text{set}}{=} (h^E)^{-1} \circ \nabla^{E*} \circ h^E$, where ∇^{E*} is the dual connection on dual bundle E^* . We say ∇^E is an Hermitian connection iff $\nabla^{E,*} = \nabla^E$.

We can now state the following significant result

Theorem 1.2.6. For every Hermitian structure h in a holomorphic bundle E with holomorphic structure $\bar{\partial}$, there exists a unique Hermitian connection ∇^E such that $\bar{\partial} = (\nabla^E)^{0,1}$, and we call this connection **Chern connection**.

Proof. We first prove the uniqueness of Chern connection. Suppose ∇^E is a Hermitian connection satisfies $\bar{\partial} = (\nabla^E)^{0,1}$, then for $s_1, s_2 \in \Gamma(E)$,

$$dh(s_1, s_2) = h(\nabla^E s_1, s_2) + h(s_1, \nabla^E s_2)$$

This reads

$$h((\nabla^E)^{1,0} s_1, s_2) = \partial h(s_1, s_2) - h(s_1, \bar{\partial} s_2)$$

Hence $\nabla^E = (\nabla^E)^{1,0} + (\nabla^E)^{0,1}$ is unique.

For existence of Chern connection, we define

$$h((\nabla^E)^{1,0} s_1, s_2) = \partial h(s_1, s_2) - h(s_1, \bar{\partial} s_2)$$

Since

$$\begin{aligned} h((\nabla^E)^{1,0}(fs_1), s_2) &= \partial h(fs_1, s_2) - h(fs_1, \bar{\partial} s_2) \\ &= \partial f h(s_1, s_2) + f \partial h(s_1, s_2) - f h(s_1, \bar{\partial} s_2) \\ &= h(\partial f s_1, s_2) + f (h((\nabla^E)^{1,0} s_1, s_2) + h(s_1, \bar{\partial} s_2)) - f h(s_1, \bar{\partial} s_2) \\ &= h(\partial f s_1 + f(\nabla^E)^{1,0} s_1, s_2) \end{aligned}$$

We deduce

$$(\nabla^E)^{1,0}(fs_1) = \partial fs_1 + f(\nabla^E)^{1,0}s_1$$

Combine above with

$$\bar{\partial}(fs_1) = \bar{\partial}fs_1 + f\bar{\partial}s_1$$

We obtain $\nabla^E \stackrel{set}{=} (\nabla^E)^{1,0} + \bar{\partial}$ is a connection satisfies its $(0,1)$ -component is coincident with $\bar{\partial}$. Then we show that the connection defined above is a Hermitian connection:

$$\begin{aligned} h(\nabla^E s_1, s_2) + h(s_1, \nabla^E s_2) &= h((\nabla^E)^{1,0}s_1, s_2) + h(\bar{\partial}s_1, s_2) \\ &\quad + h(s_1, (\nabla^E)^{1,0}s_2) + h(s_1, \bar{\partial}s_2) \\ &= \partial h(s_1, s_2) - h(s_1, \bar{\partial}s_2) + h(\bar{\partial}s_1, s_2) \\ &\quad + \overline{\partial h(s_2, s_1)} - \overline{h(s_2, \bar{\partial}s_1)} + h(s_1, \bar{\partial}s_2) \\ &= (\partial + \bar{\partial})h(s_1, s_2) \end{aligned}$$

Therefore, $\nabla^E \stackrel{set}{=} (\nabla^E)^{1,0} + \bar{\partial}$ is all we need. □

Remark 3. *Chern connection has naturality with respect to the direct sum and tensor product of holomorphic vector bundles.*

We can also deduce the existence and uniqueness of Chern connection via computing directly, before deriving corresponding formula, we need some notations. Denote matrix $(h_{\alpha\bar{\beta}}) = (h(e_\alpha, e_\beta))$ by H , then elements $h^{\alpha\bar{\beta}}$ of the inverse of H^t together with $h_{\alpha\bar{\beta}}$ satisfies

$$\begin{aligned} h^{\beta\bar{\gamma}}h_{\alpha\bar{\gamma}} &= \delta_\alpha^\beta, \quad h_{\alpha\bar{\gamma}}h^{\alpha\bar{\delta}} = \delta_\gamma^\delta, \\ \Rightarrow \quad \partial h^{\beta\bar{\gamma}} &= -h^{\beta\bar{\mu}}\partial h_{\delta\bar{\mu}}h^{\delta\bar{\gamma}}. \end{aligned}$$

Now we can state the following proposition

Proposition 1.2.7. *In local holomorphic frames $\{e_\alpha\}$ and local holomorphic coordinates $\{z_k\}$, the components of the Chern connection matrix, denoted by ω_α^β , can be computed as*

$$\omega_\alpha^\beta = \partial h_{\alpha\bar{\gamma}}h^{\beta\bar{\gamma}}. \tag{1.6}$$

For the Chern curvature, we denote it by

$$\Theta = \sum_{\alpha, \beta} \Theta_{\alpha}^{\beta} e^{\alpha} \otimes e_{\beta} = \sum_{\alpha, \beta} R_{i\bar{j}\alpha}^{\beta} dz^i \wedge d\bar{z}^j \otimes e^{\alpha} \otimes e_{\beta}. \quad (1.7)$$

The components $R_{i\bar{j}\alpha}^{\beta}$ can be computed as

$$R_{i\bar{j}\alpha}^{\beta} = -h^{\beta\bar{\gamma}} \left(\frac{\partial^2 h_{\alpha\bar{\gamma}}}{\partial z^i \partial \bar{z}^j} - h^{\delta\bar{\mu}} \frac{\partial h_{\alpha\bar{\mu}}}{\partial z^i} \frac{\partial h_{\delta\bar{\gamma}}}{\partial \bar{z}^j} \right) \quad (1.8)$$

Proof. For Chern connection ∇ , its connection matrix ω must be of type $(1, 0)$ since the frames $\{e_{\alpha}\}$ are holomorphic. Chern connection is also a Hermitian connection implies

$$dh_{\alpha\bar{\beta}} = dh(e_{\alpha}, e_{\beta}) = \omega_{\alpha}^{\gamma} h_{\gamma\bar{\beta}} + \bar{\omega}_{\beta}^{\gamma} h_{\alpha\bar{\gamma}}$$

Then consider $(1, 0)$ and $(0, 1)$ -components, we have:

$$\begin{aligned} \partial h_{\alpha\bar{\beta}} = \omega_{\alpha}^{\gamma} h_{\gamma\bar{\beta}} &\Rightarrow \omega_{\alpha}^{\gamma} = \partial h_{\alpha\bar{\beta}} h^{\gamma\bar{\beta}} \\ \bar{\partial} h_{\alpha\bar{\beta}} = \bar{\omega}_{\beta}^{\gamma} h_{\alpha\bar{\gamma}} &\Rightarrow \bar{\omega}_{\beta}^{\gamma} = \bar{\partial} h_{\alpha\bar{\beta}} h^{\alpha\bar{\gamma}} \end{aligned}$$

Thus (2.8) holds. For (2.9), by the relation $\Theta = d\omega - \omega \wedge \omega$, we get

$$\Theta_{\alpha}^{\beta} = d\omega_{\alpha}^{\beta} - \omega_{\alpha}^{\gamma} \wedge \omega_{\gamma}^{\beta}$$

(2.8) reads

$$\omega_{\alpha}^{\beta} = h^{\beta\bar{\gamma}} \frac{\partial h_{\alpha\bar{\gamma}}}{\partial z^i} dz^i$$

we have

$$\begin{aligned} d\omega_{\alpha}^{\beta} &= d \left(h^{\beta\bar{\gamma}} \frac{\partial h_{\alpha\bar{\gamma}}}{\partial z^i} \right) \wedge dz^i \\ &= \left(\left(\frac{\partial h^{\beta\bar{\gamma}}}{\partial z_j} dz_j + \frac{\partial h^{\beta\bar{\gamma}}}{\partial \bar{z}_j} d\bar{z}_j \right) \frac{\partial h_{\alpha\bar{\gamma}}}{\partial z_i} + h^{\beta\bar{\gamma}} \left(\frac{\partial^2 h_{\alpha\bar{\gamma}}}{\partial z_i \partial z_j} dz_j + \frac{\partial^2 h_{\alpha\bar{\gamma}}}{\partial z_i \partial \bar{z}_j} d\bar{z}_j \right) \right) \wedge dz_i \\ &= -h^{\beta\bar{\gamma}} \left(\frac{\partial^2 h_{\alpha\bar{\gamma}}}{\partial z^i \partial \bar{z}^j} - h^{\delta\bar{\mu}} \frac{\partial h_{\alpha\bar{\mu}}}{\partial z^i} \frac{\partial h_{\delta\bar{\gamma}}}{\partial \bar{z}^j} \right) dz^i \wedge d\bar{z}^j + h^{\delta\bar{\gamma}} \frac{\partial h_{\alpha\bar{\gamma}}}{\partial z^i} h^{\beta\bar{\mu}} \frac{\partial h_{\delta\bar{\mu}}}{\partial z^j} dz^i \wedge dz^j \end{aligned}$$

and

$$\omega_{\alpha}^{\gamma} \wedge \omega_{\gamma}^{\beta} = h^{\gamma\bar{\delta}} \frac{\partial h_{\alpha\bar{\delta}}}{\partial z^i} h^{\beta\bar{\lambda}} \frac{\partial h_{\gamma\bar{\lambda}}}{\partial z^j} dz^i \wedge dz^j$$

then

$$R_{i\bar{j}\alpha}^{\beta} = -h^{\beta\bar{\gamma}} \left(\frac{\partial^2 h_{\alpha\bar{\gamma}}}{\partial z^i \partial \bar{z}^j} - h^{\delta\bar{\mu}} \frac{\partial h_{\alpha\bar{\mu}}}{\partial z^i} \frac{\partial h_{\delta\bar{\gamma}}}{\partial \bar{z}^j} \right)$$

□

After appropriate holomorphic transformation, we obtain

Proposition 1.2.8. *Given holomorphic vector bundle (E, h) over M , a point x_0 of M and local holomorphic coordinates $(z^1, \dots, z^n)$ at x_0 , there exists a holomorphic local frame $(e_1, \dots, e_r)$ in a neighborhood U of x_0 such that*

$$h(e_\alpha(z), e_\beta(z)) = \delta_{\alpha\beta} - \sum_{j,k=1}^n R_{j\bar{k}\alpha\bar{\beta}} z_j \bar{z}_k + O(|z|^3)$$

where the $R_{j\bar{k}\alpha\bar{\beta}} = R_{i\bar{j}\alpha\bar{\beta}}^\gamma h_{\gamma\bar{\beta}}$ are the coefficients of the Chern curvature tensor $\Theta_{x_0}^E$ under the above local frame.

3 Hermitian Structure and Kähler Structure

In this section we introduce some basic concepts and facts on Hermitian structure and Kähler structure.

Definition 1.3.1. *A Hermitian metric on an almost complex manifold (M, J) is a Riemannian metric h such that $h(X, Y) = h(JX, JY), \forall X, Y \in TM$. The fundamental form of a Hermitian metric is defined by $\Omega(X, Y) \stackrel{\text{set}}{=} h(JX, Y)$.*

Remark 4. *Every almost complex manifold admits Hermitian metrics. Simply choose an arbitrary Riemannian metric g and define $h(X, Y) := g(X, Y) + g(JX, JY)$.*

Proposition 1.3.2. *The extension of the Hermitian metric h to $TM^\mathbb{C}$ by $\mathbb{C}$ -linearity satisfies*

$$\begin{cases} h(\bar{Z}, \bar{W}) = \overline{h(Z, W)}, & \forall Z, W \in TM^\mathbb{C}. \\ h(Z, \bar{Z}) > 0 & \forall Z \in TM^\mathbb{C} - \{0\}. \\ h(Z, W) = 0, & \forall Z, W \in T^{1,0}M \text{ and } \forall Z, W \in T^{0,1}M. \end{cases}$$

Conversely, each symmetric tensor on $TM^\mathbb{C}$ satisfying above properties defines a Hermitian metric by restriction to TM .

Proof. After extending to $TM^\mathbb{C}$ by $\mathbb{C}$ -linearity,

- $h(\bar{Z}, \bar{W}) = \overline{h(Z, W)}, \quad \forall Z, W \in TM^\mathbb{C}.$

Suppose $Z = (Z_1 - \sqrt{-1}JZ_1) + (Z_2 + \sqrt{-1}JZ_2)$ and $W = (W_1 - \sqrt{-1}JW_1) + (W_2 +$

$\sqrt{-1}JW_2)$, then

$$\begin{aligned}
 h(\bar{Z}, \bar{W}) &= h((Z_1 + Z_2) + \sqrt{-1}J(Z_1 - Z_2), (W_1 + W_2) + \sqrt{-1}J(W_1 - W_2)) \\
 &= h(Z_1 + Z_2, W_1 + W_2) + \sqrt{-1}h(Z_1 + Z_2, J(W_1 - W_2)) \\
 &\quad + \sqrt{-1}h(J(Z_1 - Z_2), W_1 + W_2) - h(J(Z_1 - Z_2), J(W_1 - W_2)) \\
 &= \overline{h(Z, W)}
 \end{aligned}$$

- $h(Z, \bar{Z}) > 0 \quad \forall Z \in TM^{\mathbb{C}} - \{0\}$.

Suppose $Z = (Z_1 - \sqrt{-1}JZ_1) + (Z_2 + \sqrt{-1}JZ_2)$,

$$\begin{aligned}
 h(Z, \bar{Z}) &= h((Z_1 + Z_2) - \sqrt{-1}J(Z_1 - Z_2), (Z_1 + Z_2) + \sqrt{-1}J(Z_1 - Z_2)) \\
 &= h(Z_1 + Z_2, Z_1 + Z_2) + \sqrt{-1}h(Z_1 + Z_2, J(Z_1 - Z_2)) \\
 &\quad - \sqrt{-1}h(J(Z_1 - Z_2), Z_1 + Z_2) + h(J(Z_1 - Z_2), J(Z_1 - Z_2)) \\
 &> 0
 \end{aligned}$$

- $h(Z, W) = 0, \quad \forall Z, W \in T^{1,0}M \text{ and } \forall Z, W \in T^{0,1}M$.

Suppose $Z = Z_1 - \sqrt{-1}JZ_1$ and $W = W_1 - \sqrt{-1}JW_1$,

$$\begin{aligned}
 h(Z, W) &= h(Z_1 - \sqrt{-1}JZ_1, W_1 - \sqrt{-1}JW_1) \\
 &= h(Z_1, W_1) - \sqrt{-1}h(Z_1, JW_1) - \sqrt{-1}h(JZ_1, W_1) - h(JZ_1, JW_1) \\
 &= h(Z_1, W_1) + \sqrt{-1}h(JZ_1, W_1) - \sqrt{-1}h(JZ_1, W_1) - h(Z_1, W_1) \\
 &= 0
 \end{aligned}$$

Conversely, if h is a symmetric tensor on $TM^{\mathbb{C}}$ satisfying above properties, then for $X, Y \in TM$,

$$h(X, Y) - \overline{h(X, Y)} = h(X, Y) - h(\bar{X}, \bar{Y}) = 0$$

hence h maps $TM \times TM$ to $\mathbb{R}$. Moreover,

$$\begin{aligned}
 0 &= h(X - \sqrt{-1}JX, Y - \sqrt{-1}JY) \\
 &= h(X, Y) - \sqrt{-1}h(X, JY) - \sqrt{-1}h(JX, Y) - h(JX, JY) \\
 &= (h(X, Y) - h(JX, JY)) - \sqrt{-1}(h(X, JY) + h(JX, Y)) \\
 &\Rightarrow h(X, Y) - h(JX, JY) = 0
 \end{aligned}$$

□

Proposition 1.3.3. *The formula of Lie derivative of an almost complex structure J in terms of Levi-Civita connection ∇ is given by*

$$(\mathcal{L}_X J)(Y) = (\nabla_X J)Y - \nabla_{JY} X + J\nabla_Y X$$

Proof. Actually, above proposition can be generalized. Suppose ∇ and ∇' are linear connections in vector bundles $E \rightarrow M$ and $E' \rightarrow M$, there is a unique linear connection ∇'' in the bundle $\text{Hom}(E, E')$ such that the product rule holds:

$$\nabla'_X(F\phi) = F(\nabla_X \phi) + (\nabla''_X F)\phi, \quad (1.9)$$

for all $X \in \Gamma(TM)$, $F \in \Gamma(\text{Hom}(E, E'))$ and $\phi \in \Gamma(E)$, where $F\phi \in \Gamma(E')$ is defined by $(F\phi)_x \stackrel{\text{set}}{=} F_x \phi_x$. From now on let's denote all the connections simply by ∇ . Notice that J is a section of $\text{Hom}(TM, TM)$, and we use the connection in this hom-bundle induced by the Levi-Civita connection of (M, g) . So we have

$$\begin{aligned} (\mathcal{L}_X J)(Y) &= \mathcal{L}_X(JY) - J(\mathcal{L}_X Y) \\ &= [X, JY] - J[X, Y] \\ &= \nabla_X(JY) - \nabla_{JY} X - J(\nabla_X Y - \nabla_Y X) \\ &= \nabla_X(JY) - J(\nabla_X Y) - \nabla_{JY} X + J(\nabla_Y X) \\ &= (\nabla_X J)(Y) - \nabla_{JY} X + J(\nabla_Y X) \end{aligned}$$

as wanted. In the first equal sign I use that $\mathcal{L}_X$ is a derivation, in the second one the characterization $\mathcal{L}_X = [X, \cdot]$ (when acting on vector fields, such as JY), in the third one that ∇ is torsion-free, and in the last one the characterization of the connection induced in $\text{Hom}(TM, TM)$ applied for J playing the role of F in (2.11). Also note that we did not use that J is an almost complex structure. □

Definition 1.3.4. *A Hermitian metric h on an almost complex manifold (M, J) is called a Kähler metric if J is a complex structure and the fundamental form Ω is closed, i.e.*

$$h \text{ is Kähler} \iff \begin{cases} N^J = 0 \\ d\Omega = 0 \end{cases}$$

A local real function u satisfying $\Omega = i\partial\bar{\partial}u$ is called a local Kähler potential of the metric h .

Proposition 1.3.5. *Every Hermitian metric on a 2-dimensional almost complex manifold is Kähler.*

Proof. Suppose h is a Hermitian metric on 2-dimensional almost complex manifold (M, J) . We firstly show that J is a complex structure. For any point $p \in M$, we can take a nonvanishing local vector field X around p , then $\{X, JX\}$ forms a local basis of vector fields around p since $JX = kX \Rightarrow k^2 = -1$, that is a contradiction. Specifically,

$$\begin{aligned} N^J(X, JX) &= [X, JX] + J[JX, JX] + J[X, -X] - [JX, -X] \\ &= [X, JX] + [JX, X] \\ &= 0 \end{aligned}$$

According to arbitrariness of p , one deduces N^J vanishes, Proposition 2.2 implies that J is a complex structure. The fundamental form Ω is closed, since $d\Omega$ is a 3-form, while there are no non-trivial 3-forms in 2-dimensional manifolds. □

Lemma 1.3.6. *Let h be a Hermitian metric on an almost complex manifold (M, J) , with Levi-Civita covariant derivative ∇ . Then J is integrable if and only if*

$$(\nabla_{JX}J)Y = J(\nabla_XJ)Y, \quad \forall X, Y \in TM.$$

Proof. Suppose $\forall X, Y \in TM$, $(\nabla_{JX}J)Y = J(\nabla_XJ)Y$ holds, we prove $\forall X, Y \in TM$, $N^J(X, Y) = 0$, where N^J is the Nijenhuis tensor. Extend $X, Y \in TM$ to vector fields on neighbourhoods, since ∇ torsion free, we have

$$\begin{aligned} N^J(X, Y) &= [X, Y] + J[JX, Y] + J[X, JY] - [JX, JY] \\ &= (\nabla_XY - \nabla_YX) + J(\nabla_{JX}Y - \nabla_YJX) \\ &\quad + J(\nabla_XJY - \nabla_{JY}X) - (\nabla_{JX}JY - \nabla_{JY}JX) \\ &= J(\nabla_XJY - J\nabla_XY) - J(\nabla_YJX - J\nabla_YX) \\ &\quad - (\nabla_{JX}JY - J\nabla_{JX}Y) + (\nabla_{JY}JX - J\nabla_{JY}X) \\ &= J(\nabla_XJ)Y - J(\nabla_YJ)X - (\nabla_{JX}J)Y + (\nabla_{JY}J)X \\ &= (J(\nabla_XJ)Y - (\nabla_{JX}J)Y) - (J(\nabla_YJ)X - (\nabla_{JY}J)X) \\ &= 0 \end{aligned} \tag{1.10}$$

Thus J is integrable.

Conversely, suppose J is integrable, we need to prove $\forall X, Y \in TM$,

$$(\nabla_{JX}J)Y = J(\nabla_XJ)Y$$

holds. Equivalently, define a new tensor

$$A(X, Y, Z) \stackrel{\text{set}}{=} h(J(\nabla_XJ)Y - (\nabla_{JX}J)Y, Z)$$

We enumerate several properties of tensor A , which is sufficient to illustrate that $A \equiv 0$.

- $A(X, Y, Z) = A(Y, X, Z)$

(2.12) reads

$$\begin{aligned} A(X, Y, Z) - A(Y, X, Z) &= h(J(\nabla_XJ)Y - (\nabla_{JX}J)Y, Z) \\ &\quad - h(J(\nabla_YJ)X - (\nabla_{JY}J)X, Z) \\ &= H(N^J(X, Y), Z) \\ &= 0 \end{aligned}$$

- $A(X, Y, Z) = -A(X, Z, Y)$

Since ∇ is a Levi-Civita connection, we have

$$\begin{aligned} h(J(\nabla_XJ)Y, Z) + h(J(\nabla_XJ)Z, Y) &= (-h(\nabla_XJY, JZ) + h(\nabla_XY, Z)) \\ &\quad + (-h(\nabla_XJZ, JY) + h(\nabla_XZ, Y)) \\ &= -(h(\nabla_XJY, JZ) + h(\nabla_XJZ, JY)) \\ &\quad + (h(\nabla_XY, Z) + h(\nabla_XZ, Y)) \\ &= -Xh(JY, JZ) + Xh(Y, Z) \\ &= 0 \end{aligned} \tag{1.11}$$

and

$$\begin{aligned}
 h((\nabla_{JX}J)Y, Z) + h((\nabla_{JX}J)Z, Y) &= (h(\nabla_{JX}JY, Z) + h(\nabla_{JX}Y, JZ)) \\
 &\quad + (h(\nabla_{JX}JZ, Y) + h(\nabla_{JX}Z, JY)) \\
 &= (h(\nabla_{JX}JY, Z) + h(\nabla_{JX}Z, JY)) \\
 &\quad + (h(\nabla_{JX}Y, JZ) + h(\nabla_{JX}JZ, Y)) \\
 &= JXh(JY, Z) + JXh(Y, JZ) \\
 &= 0
 \end{aligned} \tag{1.12}$$

Combine (2.13) with (2.14) we obtain $A(X, Y, Z) = -A(X, Z, Y)$.

The symmetry of tensor A with respect to its arguments implies

$$\begin{aligned}
 A(X, Y, Z) &= -A(X, Z, Y) = -A(Z, X, Y) = A(Z, Y, X) = A(Y, Z, X) \\
 A(X, Y, Z) &= A(Y, X, Z) = -A(Y, Z, X) \\
 &\Rightarrow A \equiv 0.
 \end{aligned}$$

Hence

$$(\nabla_{JX}J)Y = J(\nabla_XJ)Y$$

holds for all $X, Y \in TM$.

□

Theorem 1.3.7. *A Hermitian metric h on an almost complex manifold (M, J) is a Kähler metric iff J is parallel with respect to the Levi-Civita connection of h .*

Proof. It suffices to prove if h is a Kähler metric, then J is parallel with respect to Levi-Civita connection ∇ of h . For this purpose, we define a new tensor:

$$B(X, Y, Z) \stackrel{\text{set}}{=} h((\nabla_XJ)Y, Z)$$

We enumerate several properties of tensor B , which is sufficient to illustrate that $B(X, Y, JZ) = 0$ for any $X, Y, Z \in TM$.

- $B(X, Y, JZ) = B(X, JY, Z)$.

$$\begin{aligned}
 B(X, Y, JZ) - B(X, JY, Z) &= (h(\nabla_X JY, JZ) - h(J(\nabla_X Y), JZ)) \\
 &\quad - (h(-\nabla_X Y, Z) + h(-J(\nabla_X JY), Z)) \\
 &= (h(\nabla_X JY, JZ) + h(J(\nabla_X JY), Z)) \\
 &\quad + (h(\nabla_X Y, Z) - h(J(\nabla_X Y), JZ)) \\
 &= 0
 \end{aligned}$$

- $B(X, Y, JZ) = -B(JX, Y, Z)$.

Since h is Kähler metric, Lemma 2.2 implies

$$B(X, Y, JZ) = -h(J(\nabla_X J)Y, Z) = -h((\nabla_{JX} J)Y, Z) = -B(JX, Y, Z)$$

- $B(JX, Y, Z) = -B(X, JY, Z)$.

$$\begin{cases} B(X, Y, JZ) = B(X, JY, Z) \\ B(X, Y, JZ) = -B(JX, Y, Z) \end{cases} \Rightarrow B(X, JY, Z) = -B(JX, Y, Z)$$

- $B(X, Y, JZ) + B(Y, JZ, X) + B(JZ, X, Y) = 0$.

The left handside can be written as

$$\begin{aligned}
 \text{LHS} &= (h(\nabla_X JY, JZ) - h(J(\nabla_X Y), JZ)) \\
 &\quad - (h(\nabla_Y Z, X) + h(J(\nabla_Y JZ), X)) \\
 &\quad + (h(\nabla_{JZ} JX, Y) - h(J(\nabla_{JZ} X), Y))
 \end{aligned} \tag{1.13}$$

Since h is Kähler metric, fundamental form Ω is closed, then

$$\begin{aligned}
 0 &= d\Omega(X, Y, JZ) \\
 &= X(h(Y, Z)) - Y(h(X, Z)) + JZ(h(JX, Y)) \\
 &\quad - h(\nabla_X Y - \nabla_Y X, Z) - h(\nabla_X JZ - \nabla_{JZ} X, JY) + h(\nabla_Y JZ - \nabla_{JZ} Y, JX) \\
 &= h(Y, \nabla_X Z) - h(X, \nabla_Y Z) + h(\nabla_{JZ} JX, Y) \\
 &\quad - h(\nabla_X JZ, JY) + h(\nabla_{JZ} X, JY) + h(\nabla_Y JZ, JX)
 \end{aligned} \tag{1.14}$$

Combine (2.15) and (2.16), we have

$$\begin{aligned}
 \text{LHS} &= \text{LHS} - d\Omega(X, Y, JZ) \\
 &= Xh(JY, JZ) - Xh(Y, Z) \\
 &= 0
 \end{aligned}$$

- $B(X, JY, Z) + B(JY, Z, X) + B(Z, X, JY) = 0$.

Proof of this identity is similar to previous one, except modifying computation in (2.15) to $0 = d\Omega(X, JY, Z)$.

The symmetry of tensor B with respect to its arguments implies

$$\begin{aligned}
 0 &= (B(X, Y, JZ) + B(Y, JZ, X) + B(JZ, X, Y)) \\
 &\quad + (B(X, JY, Z) + B(JY, Z, X) + B(Z, X, JY)) \\
 &= 2B(X, Y, JZ) + (B(Y, JZ, X) + B(JY, Z, X)) \\
 &\quad + (B(JZ, X, Y) + B(Z, X, JY)) \\
 &= 2B(X, Y, JZ)
 \end{aligned}$$

That is, J is parallel with respect to the Levi-Civita connection ∇ of h .

□

In the reminder of this section, we make some comparison of the Levi-Civita connection and Chern connection. To do so, we firstly express the $\bar{\partial}$ operator on a Hermitian manifold in terms of Levi-Civita connection ∇ . Remember one may identify TM with a complex vector bundle via complex structure J , more specifically, one may identify the element $\sqrt{-1}X$ with JX .

Lemma 1.3.8. *For every section Y of the complex vector bundle TM , the $\bar{\partial}$ -operator, as TM -valued $(0, 1)$ -form is given by*

$$\bar{\partial}^\nabla Y(X) = \frac{1}{2} (\nabla_X Y + J\nabla_{JX} Y - J(\nabla_Y J)X), \quad (1.15)$$

where ∇ denotes the Levi-Civita connection of any Hermitian metric h on M .

Proof. Firstly inspect that (2.17) actually satisfies Leibniz rule:

$$\begin{aligned}
 \bar{\partial}^\nabla fY(X) &= \frac{1}{2}(X(f)Y + f\nabla_X Y) + \frac{1}{2}J(JX(f)Y + f\nabla_{JX} Y) \\
 &\quad - \frac{1}{2}J(\nabla_{fY}(JX) - J(\nabla_{fY} X)) \\
 &= \frac{1}{2}(f\nabla_X Y + fJ(\nabla_{JX} Y) - J\nabla_{fY}(JX) - (\nabla_{fY} X)) \\
 &\quad + \frac{1}{2}(X(f)Y + JX(f)JY) \\
 &= f\bar{\partial}^\nabla Y(X) + \bar{\partial}f(X)Y
 \end{aligned}$$

where in the last step we used

$$\frac{1}{2}(X(f)Y + JX(f)JY) = \frac{1}{2}(X(f)Y + \sqrt{-1}JX(f)Y) = \bar{\partial}f(X)Y$$

We state a fact: a vector field X is a holomorphic section of $TM \Rightarrow$ it is real holomorphic $\Rightarrow \mathcal{L}_X J = 0$. Then Proposition (2.7) reads for all $X, Y \in \Gamma(TM)$

$$\begin{aligned}
 0 &= (\mathcal{L}_X J)(Y) = (\nabla_X J)Y - \nabla_{JY} X + J\nabla_Y X \\
 &= J(\nabla_Y X - J((\nabla_X J)Y) + J(\nabla_{JY} X)) \\
 &= 2J\bar{\partial}^\nabla X(Y)
 \end{aligned}$$

Thus $\bar{\partial}^\nabla X$ vanishes for every holomorphic section X , this implies that $\bar{\partial}^\nabla$ coincides with $\bar{\partial}$. □

Theorem 1.3.9. *The Chern connection coincides with the Levi-Civita connection if and only if h is Kähler.*

Proof. • proof of **only if** direction.

• proof of **if** direction. □

4 Hodge and Dolbeault Theory

We firstly review some basic facts on Hodge theory. Suppose (M, g) is a compact, oriented n -dimensional manifold. We denote the space of smooth k -forms by $\Omega^k M$ and

by $\mathcal{Z}^k(M)$ the space of closed k -forms on M . Relation $d\Omega^{k-1}M \subset \mathcal{Z}^k M$ permits us to define the De Rham cohomology groups by

$$H_{dR}^k(M) := \frac{\mathcal{Z}^k M}{d\Omega^{k-1}M}.$$

The De Rham theorem says that the k^{th} cohomology group of M is naturally isomorphic to the k^{th} De Rham cohomology group, i.e.

$$H^k(M) \simeq H_{dR}^k(M).$$

We now denote by $\mathcal{H}^k(M)$ the space of harmonic k -forms on M , i.e. forms in the kernel of the Laplace operator:

$$\mathcal{H}^k(M) := \{\omega \in \Omega^k M \mid \Delta\omega = 0\}.$$

Then we have

Theorem 1.4.1 (Hodge decomposition theorem). *The space of k -forms decomposes as a direct sum*

$$\Omega^k M = \mathcal{H}^k(M) \oplus \delta\Omega^{k+1}M \oplus d\Omega^{k-1}M.$$

Here, three spaces are orthogonal with respect to the L^2 inner product on $\Omega^k M$ given by:

$$(\alpha, \beta) \stackrel{\text{set}}{=} \int_M g(\alpha, \beta) \omega_g$$

Theorem 1.4.2 (Hodge isomorphism). *The natural map $f : \mathcal{H}^k(M) \rightarrow H_{dR}^k(M)$ given by $\omega \mapsto [\omega]$ is an isomorphism.*

Proof. • f is well-defined.

This obvious since harmonic forms are closed.

• f is injective.

For any $\omega \in \ker f$, $f(\omega) = 0$ means there exists η , s.t. $\omega = d\eta$, which implies ω is harmonic and exact. While $\mathcal{H}^k(M)$ and $d\Omega^{k-1}M$ are orthogonal, we have $\omega = 0$.

- f is surjective.

For any de Rham cohomology class c , there exists a closed form ω s.t. $[\omega] = c$. The Hodge decomposition of ω writes $\omega = d\omega' + \omega^H$, then

$$f(\omega^H) = [\omega^H] = [d\omega' + \omega^H] = c$$

□

The dimension $b_k(M) \stackrel{set}{=} \dim(H_{DR}^k(M))$ is called the k^{th} Betti number of M and is a topological invariant in view of De Rham's theorem.

Theorem 1.4.3 (Poincaré duality). *The spaces $\mathcal{H}^k(M)$ and $\mathcal{H}^{n-k}(M)$ are isomorphic. In particular $b_k(M) = b_{n-k}(M)$ for every compact n -dimensional manifold M .*

Proof. Since Hodge $*$ -operator commutes with Hodge Laplace operator, and Hodge $*$ -operator satisfies $*^2 = (-1)^{k(n-k)}$, the isomorphism is simply given by the Hodge $*$ -operator which sends harmonic k -forms to harmonic $n - k$ -forms.

□

Before introducing cohomology theory on compact Hermitian manifold (M^{2m}, h, J) , we recall some facts on Laplace operators on Hermitian manifold.

Corresponding to the decomposition $d = \partial + \bar{\partial}$ we have the decomposition $\delta = \partial^* + \bar{\partial}^*$, where (The Hodge $*$ -operator maps (p, q) -forms to $(m - q, m - p)$ -forms)

$$\begin{aligned} \partial^* : \Gamma(\Lambda^{p,q}M) &\rightarrow \Gamma(\Lambda^{p-1,q}M), & \partial^* &:= - * \bar{\partial} * \\ \bar{\partial}^* : \Gamma(\Lambda^{p,q}M) &\rightarrow \Gamma(\Lambda^{p,q-1}M), & \bar{\partial}^* &:= - * \partial * . \end{aligned}$$

Notice that ∂^* and $\bar{\partial}^*$ are formal adjoints of ∂ and $\bar{\partial}$ with respect to the Hermitian product H on complex forms given by

$$H(\omega, \tau) \stackrel{set}{=} \langle \omega, \bar{\tau} \rangle. \tag{1.16}$$

In the right handside of (2.18), $\langle \cdot, \cdot \rangle$ is defined as

$$\langle \omega, \tau \rangle \stackrel{set}{=} \frac{1}{k!} g(\varphi(\omega), \varphi(\tau)),$$

where

$$\varphi(e_1 \wedge \dots \wedge e_k) = \sum_{\sigma \in S_k} \varepsilon(\sigma) e_{\sigma_1} \otimes \dots \otimes e_{\sigma_k}.$$

Define the Laplace operators

$$\Delta^\partial := \partial\partial^* + \partial^*\partial \quad \text{and} \quad \Delta^{\bar{\partial}} := \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}.$$

One of the most important features of Kähler metrics is that these new Laplace operators are essentially the same as the usual one, i.e.

Lemma 1.4.4. *On Kähler manifolds one has $\Delta = 2\Delta^\partial = 2\Delta^{\bar{\partial}}$.*

We need to make some preparations to prove this lemma.

Denote the fundamental form on Hermitian manifold M by Ω , and identify a $(1, 0)$ -vector with a $(0, 1)$ -form via the metric map, we can define several operators as following:

$$\begin{aligned} L : \Lambda^k M &\rightarrow \Lambda^{k+2} M, \\ \omega &\mapsto \Omega \wedge \omega = \frac{1}{2} \sum_i e_i \wedge J e_i \wedge \omega, \\ \Lambda : \Lambda^{k+2} M &\rightarrow \Lambda^k M, \\ \omega &\mapsto \frac{1}{2} \sum_i J e_i \lrcorner e_i \lrcorner \omega. \end{aligned}$$

where Λ is adjoint operator of L , and these natural operators can be extended to complex-valued forms by $\mathbb{C}$ -linearity. Straightforward computation shows

$$[X \lrcorner, \Lambda] = 0, \quad [X \lrcorner, L] = JX \wedge. \quad (1.17)$$

Rigorously, we check for 1-form ω :

$$\begin{aligned} [X \lrcorner, L]\omega &= \frac{1}{2} X \lrcorner (e_i \wedge J e_i \wedge \omega) - \frac{1}{2} e_i \wedge J e_i \wedge (X \lrcorner \omega) \\ &= \frac{1}{2} (e_i(X) J e_i \wedge \omega - J e_i(X) e_i \wedge \omega + \omega(X) e_i \wedge J e_i) \\ &\quad - \frac{1}{2} \omega(X) e_i \wedge J e_i \\ &= (JX) \wedge \omega \end{aligned}$$

Suppose that M is Kähler, and define twisted differential operator d^c :

$$\begin{aligned} d^c : \Gamma(\Lambda^k M) &\rightarrow \Gamma(\Lambda^{k+1} M), \\ \omega &\mapsto \sum_i J e_i \wedge \nabla_{e_i} \omega. \end{aligned}$$

whose formal adjoint operator $\delta^c = - * d^c *$ can be written as

$$\begin{aligned} \delta^c : \Gamma(\Lambda^{k+1} M) &\rightarrow \Gamma(\Lambda^k M), \\ \omega &\mapsto - \sum_i J e_i \lrcorner \nabla_{e_i} \omega. \end{aligned}$$

We have

Lemma 1.4.5. *On Kähler manifold, the relations*

$$\begin{aligned} 1) \quad [L, \delta] &= d^c, \quad [L, d] = 0, \\ 2) \quad [\Lambda, d] &= -\delta^c, \quad [\Lambda, \delta] = 0. \end{aligned}$$

hold.

Proof. We use (2.19) and J is parallel,

$$\begin{aligned} [L, \delta] &= -[L, e_i \lrcorner \nabla_{e_i}] = -[L, e_i] \lrcorner \nabla_{e_i} = J e_i \wedge \nabla_{e_i} = d^c. \\ [L, d] &= [L, e_i \wedge \nabla_{e_i}] = 0 \end{aligned}$$

To derive 2), we only need to take the adjoint operator of L, d, d^c .

□

Now we can prove lemma (2.4).

Proof. From the fact that $\Lambda^{p,q} M$ are preserved by the covariant derivative follows easily

$$\partial = \sum_j \frac{1}{2} (e_j + \sqrt{-1} J e_j) \wedge \nabla_{e_j} \quad \text{and} \quad \bar{\partial} = \sum_j \frac{1}{2} (e_j - \sqrt{-1} J e_j) \wedge \nabla_{e_j}.$$

From the definition of d^c we then get

$$d^c = \sqrt{-1}(\bar{\partial} - \partial),$$

and take the adjoint operator of $\partial, \bar{\partial}, d^c$

$$\delta^c = \sqrt{-1}(\partial^* - \bar{\partial}^*).$$

Applying lemma (2.5) to a (p, q) -form and projecting onto $\Lambda^{p\pm 1, q}M$ and $\Lambda^{p, q\pm 1}M$ then yields

$$[L, \partial^*] = \sqrt{-1}\bar{\partial}, \quad [L, \bar{\partial}^*] = -\sqrt{-1}\partial, \quad [L, \partial] = 0, \quad [L, \bar{\partial}] = 0, \quad (1.18)$$

$$[\Lambda, \partial] = \sqrt{-1}\bar{\partial}^*, \quad [\Lambda, \bar{\partial}] = -\sqrt{-1}\partial^*, \quad [\Lambda, \partial^*] = 0, \quad [\Lambda, \bar{\partial}^*] = 0. \quad (1.19)$$

Therefore, the relation $\bar{\partial}^2 = 0$ together with (2.21) reads

$$-\sqrt{-1}(\bar{\partial}\partial^* + \partial^*\bar{\partial}) = \bar{\partial}[\Lambda, \bar{\partial}] + [\Lambda, \bar{\partial}]\bar{\partial} = \bar{\partial}\Lambda\bar{\partial} - \bar{\partial}^2\Lambda + \Lambda\bar{\partial}^2 - \bar{\partial}\Lambda\bar{\partial} = 0$$

Similarly

$$-\sqrt{-1}(\partial\bar{\partial}^* + \bar{\partial}^*\partial) = [L, \bar{\partial}^*]\bar{\partial}^* + \bar{\partial}^*[L, \bar{\partial}^*] = L(\bar{\partial}^*)^2 - \bar{\partial}^*L\bar{\partial}^* + \bar{\partial}^*L\bar{\partial}^* - (\bar{\partial}^*)^2L = 0.$$

Hence

$$\begin{aligned} \Delta &= (\partial + \bar{\partial})(\partial^* + \bar{\partial}^*) + (\partial^* + \bar{\partial}^*)(\partial + \bar{\partial}) \\ &= (\partial\partial^* + \partial^*\partial + \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}) \\ &= \Delta^\partial + \Delta^{\bar{\partial}}. \end{aligned}$$

Eventually, it remains to show the equality $\Delta^\partial = \Delta^{\bar{\partial}}$. (2.21) reads

$$\begin{aligned} -\sqrt{-1}\Delta^\partial &= -\sqrt{-1}(\partial\partial^* + \partial^*\partial) \\ &= \partial[\Lambda, \bar{\partial}] + [\Lambda, \bar{\partial}]\partial \\ &= \partial\Lambda\bar{\partial} - \partial\bar{\partial}\Lambda + \Lambda\bar{\partial}\partial - \bar{\partial}\Lambda\partial \\ &= \partial\Lambda\bar{\partial} + \bar{\partial}\partial\Lambda - \Lambda\partial\bar{\partial} - \bar{\partial}\Lambda\partial \\ &= [\partial, \Lambda]\bar{\partial} + \bar{\partial}[\partial, \Lambda] \\ &= -\sqrt{-1}\bar{\partial}^*\bar{\partial} - \sqrt{-1}\bar{\partial}\bar{\partial}^* \\ &= -\sqrt{-1}\Delta^{\bar{\partial}}. \end{aligned}$$

□

Now suppose (M^{2m}, h, J) be a compact Hermitian manifold, we consider the **Dolbeault operator** $\bar{\partial}$ acting on the spaces of (p, q) -forms $\Omega^{p,q}M \stackrel{\text{set}}{=} \Gamma(\Lambda^{p,q}M) \subset \Omega^{p+q}M$. Similarly, let $\mathcal{Z}^{p,q}M$ denote the space of $\bar{\partial}$ -closed (p, q) -forms, and relation $\bar{\partial}\Omega^{p,q-1}M \subset \mathcal{Z}^{p,q}M$ permits us to define the Dolbeault cohomology groups

$$H^{p,q}M := \frac{\mathcal{Z}^{p,q}M}{\bar{\partial}\Omega^{p,q-1}M}.$$

In contrast to De Rham cohomology, the Dolbeault cohomology is no longer a topological invariant of the manifold, since it strongly depends on the complex structure J .

We define the space $\mathcal{H}^{p,q}M$ of $\bar{\partial}$ -harmonic (p, q) -forms on M by

$$\mathcal{H}^{p,q}M := \left\{ \omega \in \Omega^{p,q}M \mid \Delta^{\bar{\partial}}\omega = 0 \right\}.$$

Then immediately

Lemma 1.4.6. *A form $\omega \in \Omega^{p,q}M$ is $\bar{\partial}$ -harmonic if and only if $\bar{\partial}\omega = 0$ and $\bar{\partial}^*\omega = 0$.*

Proof. One direction is obvious. Conversely, suppose ω is $\bar{\partial}$ -harmonic, then

$$0 = \int_M h(\Delta^{\bar{\partial}}\omega, \omega) dv = \int_M h((\bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial})\omega, \omega) dv = \int_M |\bar{\partial}\omega|^2 + |\bar{\partial}^*\omega|^2 dv$$

which implies $\bar{\partial}\omega = 0$ and $\bar{\partial}^*\omega = 0$. □

Then we have

Theorem 1.4.7 (Dolbeault Decomposition Theorem). *The space of (p, q) -forms decomposes as a direct sum*

$$\Omega^{p,q}M = \mathcal{H}^{p,q}M \oplus \bar{\partial}^*\Omega^{p,q+1}M \oplus \bar{\partial}\Omega^{p,q-1}M.$$

Lemma 2.4 shows that the three spaces above are orthogonal with respect to the global Hermitian inner product

$$(\cdot, \cdot) := \int_M h(\cdot, \cdot) dv$$

on $\Omega^{p,q}M$.

This theorem shows that every (p, q) -form ω on M can be uniquely decomposed as

$$\omega = \bar{\partial}\omega' + \bar{\partial}^*\omega'' + \omega^H,$$

where $\omega' \in \Omega^{p,q-1}M$, $\omega'' \in \Omega^{p,q+1}M$, and $\omega^H \in \mathcal{H}^{p,q}M$. Since the second summand in the Dolbeault decomposition of ω vanishes if and only if $\bar{\partial}\omega = 0$, specializing for $q = 0$ yields

Proposition 1.4.8. *A $(p, 0)$ -form on a compact Hermitian manifold is holomorphic if and only if it is $\bar{\partial}$ -harmonic.*

Proof. For one direction, suppose $(p, 0)$ -form ω is holomorphic, then

$$\begin{aligned}\Delta^{\bar{\partial}}\omega &= (\bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}) (\bar{\partial}\omega' + \bar{\partial}^*\omega'' + \omega^H) \\ &= \bar{\partial}\bar{\partial}^*\bar{\partial}\omega' + \bar{\partial}^*\bar{\partial}\bar{\partial}^*\omega'' \\ &= 0\end{aligned}$$

Conversely, suppose $(p, 0)$ -form ω is $\bar{\partial}$ -harmonic, then Lemma (2.6) implies ω is holomorphic. □

Theorem 1.4.9 (Dolbeault isomorphism). *The map $f : \mathcal{H}^{p,q}M \rightarrow H^{p,q}M$ given by $\omega \mapsto [\omega]$ is an isomorphism.*

Proof. • f is well-defined.

This is obvious since $\bar{\partial}$ -harmonic forms are $\bar{\partial}$ -closed.

- f is injective.

For any $\omega \in \ker f$, $f(\omega) = 0$ means there exists η , s.t. $\omega = \bar{\partial}\eta$, which implies ω is $\bar{\partial}$ -harmonic and $\bar{\partial}$ -exact. While $\mathcal{H}^{p,q}(M)$ and $\bar{\partial}\Omega^{p,q-1}M$ are orthogonal, we have $\omega = 0$.

- f is surjective.

For any Dolbeault cohomology class c , there exists a $\bar{\partial}$ -closed form ω s.t. $[\omega] = c$.

The Hodge decomposition of ω writes $\omega = \bar{\partial}\omega' + \omega^H$, then

$$f(\omega^H) = [\omega^H] = [\bar{\partial}\omega' + \omega^H] = c$$

□

□

We denote by $h^{p,q}$ the complex dimension of $H^{p,q}(M)$. These are the **Hodge numbers** associated to the complex structure J of M . Now we can state the following theorem:

Theorem 1.4.10 (Serre duality). *The spaces $\mathcal{H}^{p,q}M$ and $\mathcal{H}^{m-p,m-q}M$ are isomorphic. In particular $h^{p,q} = h^{m-p,m-q}$.*

Proof. Consider the composition of the Hodge $*$ -operator with the complex conjugation

$$\bar{*} : \Omega^{p,q}M \rightarrow \Omega^{m-p,m-q}M, \quad \bar{*}(\omega) := *\bar{\omega}.$$

We have

$$\begin{aligned} \bar{*}\Delta^{\bar{\partial}}(\omega) &= \overline{*(\partial\bar{\partial}^* + \bar{\partial}^*\partial)\omega} \\ &= *(\partial\bar{\partial}^* + \bar{\partial}^*\partial)\bar{\omega} \\ &= -*(\partial*\bar{\partial}^* + *\bar{\partial}^*\partial)\bar{\omega} \\ &= \bar{\partial}^*\bar{\partial}(\bar{*}\omega) - *\bar{\partial}^*\partial\bar{\omega} \\ &= \bar{\partial}^*\bar{\partial}(\bar{*}\omega) - \bar{\partial}^*\partial*\bar{\omega} \\ &= \bar{\partial}^*\bar{\partial}(\bar{*}\omega) + \bar{\partial}\bar{\partial}^*(\bar{*}\omega) \\ &= \Delta^{\bar{\partial}}(\bar{*}\omega). \end{aligned}$$

This shows that $\bar{*}$ is a ($\mathbb{C}$ -anti-linear isomorphism from $\mathcal{H}^{p,q}M$ to $\mathcal{H}^{m-p,m-q}M$. Moreover, operator $\bar{*}$ satisfies exhibiting reflective characteristics, hence $\bar{*}$ is an isomorphism between the spaces $\mathcal{H}^{p,q}M$ and $\mathcal{H}^{m-p,m-q}M$.

□

Additionally, we investigate the relation between Hodge and Betti numbers on compact Kähler manifolds. Associated Result can be stated as

Proposition 1.4.11. *On compact Kähler manifolds, the following relations between Betti and Hodge numbers hold*

$$b_k = \sum_{p+q=k} h^{p,q}, \quad h^{p,q} = h^{q,p}, \quad h^{p,p} \geq 1 \quad \forall 0 \leq p \leq m.$$

In particular, all Betti numbers of odd order are even and all Betti numbers of even order are non-zero.

Proof. Firstly, the fact that $\Delta = 2\Delta^{\bar{\partial}}$ shows that $\mathcal{H}^{p,q}M \subset \mathcal{H}^{p+q}M$. Secondly, since $\Delta^{\bar{\partial}}$ leaves the spaces $\Omega^{p,q}M$ invariant, we deduce that Δ has the same property, thus proving that the components of a harmonic form in its type decomposition are all harmonic forms.

This shows that

$$\begin{aligned}
 \mathcal{H}^k M &= \{\omega \in \oplus_{p+q=k} \Lambda^{p,q} M \mid \Delta \omega = 0\} \\
 &= \left\{ \omega \in \oplus_{p+q=k} \Lambda^{p,q} M \mid \Delta^{\bar{\partial}} \omega = 0 \right\} \\
 &= \oplus_{p+q=k} \mathcal{H}^{p,q} M.
 \end{aligned}$$

Hence $b_k = \sum_{p+q=k} h^{p,q}$. Moreover, the fact that $\Delta = 2\Delta^{\bar{\partial}}$ shows that $\Delta^{\bar{\partial}}$ commutes with complex conjugation (in the general case we only have that $\overline{\Delta^{\bar{\partial}} \alpha} = \Delta^{\partial} \bar{\alpha}$), so the complex conjugation defines an isomorphism between the spaces $\mathcal{H}^{p,q} M$ and $\mathcal{H}^{q,p} M$, thus $h^{p,q} = h^{q,p}$.

Consider now the fundamental form $\Omega \in \Omega^{1,1} M$. Notice exterior power $\Omega^p \in \Omega^{p,p} M$ is a non-zero multiple of Ω . Moreover, Ω is parallel implies that Ω^p is parallel, which means Ω^p is harmonic, we deduce that $h^{p,p} \geq 1$.

□

An application of Proposition (2.10) is

Proposition 1.4.12. *The complex manifold $S^1 \times S^{2k+1}$ carries no Kähler metric for $k \geq 1$.*

Proof. If $S^1 \times S^{2k+1}$ carries a Kähler metric, then Proposition (2.10) reads $H_{dR}^2(S^1 \times S^{2k+1})$ is nontrivial. Directly computation shows $H_{dR}^2(S^1 \times S^{2k+1}) = 0$, that is a contradiction.

□

Chapter 2 Differential Geometry of Complex Vector Bundles

1 Flat Bundles and Flat Connections

Let E be a smooth complex vector bundle over a real manifold M . A **flat structure** in E is given by an open cover $\{U, s_U\}$ with local frame fields such that the transition functions g_{VU} are all constant matrixs in $\text{GL}(r, \mathbb{C})$. On the other hand, a connection D is said to be **flat** if its curvature R vanishes.

Now consider the relation between flat structures and flat connections.

- A flat vector bundle E admits a natural flat connection D .

If the flat structure is given by $\{U, s_U\}$, define D as follows:

$$Ds_U = 0$$

Since the transition functions g_{VU} are all constants matrixs, combine with transformation formula

$$\omega_U = g_{VU}^{-1} \omega_V g_{VU} + g_{VU}^{-1} dg_{VU} \quad (2.1)$$

(Where ω_U, ω_V are connection matrixs of connection D) We obtain $\omega_V = 0$, which means $Ds_V = 0$. Thus D is well defined. Obviously, the curvature of D vanishes.

- A vector bundle E with a flat connection D admits a natural flat structure $\{U, s_U\}$. Suppose vector bundle E with a flat connection D , i.e. curvature vanishes, thus we can choose two local parallel frame fields s and s' , then $s = s'g$ for some $g \in \text{GL}(r, \mathbb{C})$ since both connection matrixs vanish in (1.8). Therefore, if $Ds_U = Ds_V = 0$,

the corresponding transition function g_{VU} is constant matrix, which implies a flat connection D gives rise to a flat structure.

In summary, we have established the following

Theorem 2.1.1. *For a complex vector bundle E of rank r over M , the following two conditions are equivalent:*

- 1) *E admits a flat structure.*
- 2) *E admits a flat connection D .*

Our target is to deduce more results on bundle E through the 'flat' condition. Let E be a vector bundle with a flat connection D , denote the fundamental group of M with reference point $x_0 \in M$ by π_1 . Since the connection is flat, the parallel displacement along a closed curve c starting at x_0 depends only on the homotopy class of curve c , thus the parallel displacement gives rise to a representation:

$$\rho : \pi_1 \rightarrow \text{GL}(r, \mathbb{C}) \quad (2.2)$$

Actually, the image of ρ is the holonomy group of D . We have illustrated that a flat connection on a bundle E can induce a representation of fundamental group of base manifold.

Before we consider the converse progress, we need a proposition. Recall for a covering space $p : \tilde{M} \rightarrow M$, a covering transformation is a diffeomorphism $\phi : \tilde{M} \rightarrow \tilde{M}$ such that $p \circ \phi = p$ and all covering transformations form a group under composition, denoted by $G(\tilde{M})$. A covering space $p : \tilde{M} \rightarrow M$ is called normal if for each $x \in M$ and each pair of lifts $\tilde{x}, \tilde{x}'$ of x there is a covering transformation taking $\tilde{x}$ to $\tilde{x}'$

Proposition 2.1.2. *Let $p : (\tilde{M}, \tilde{x}_0) \rightarrow (M, x_0)$ be a path-connected covering space of the path-connected, locally path-connected space M , and let H be the subgroup $p_*(\pi_1(\tilde{M}, \tilde{x}_0)) \subset \pi_1(M, x_0)$. Then*

- 1) *This covering space is normal iff H is a normal subgroup of $\pi_1(M, x_0)$.*
- 2) *$G(\tilde{M})$ is isomorphic to the quotient $N(H)/H$ where $N(H)$ is the normalizer of H in $\pi_1(M, x_0)$.*

In particular, $G(\tilde{M})$ is isomorphic to $\pi_1(M, x_0)/H$ if $\tilde{M}$ is a normal covering. Hence for the universal cover $\tilde{M} \rightarrow M$ we have $G(\tilde{M}) \cong \pi_1(M)$.

Now suppose representation (2.2) is given, construct a vector bundle E by setting

$$E = \tilde{M} \times_{\rho} \mathbb{C}^r$$

Where $\tilde{M}$ is the universal cover of M and $\tilde{M} \times_{\rho} \mathbb{C}^r$ denotes the quotient of $\tilde{M} \times \mathbb{C}^r$ by the action of π_1 , which can be written as

$$\begin{aligned} (\tilde{M} \times \mathbb{C}^r) \times \pi_1 &\rightarrow \tilde{M} \times \mathbb{C}^r \\ (x, v)\gamma &\mapsto (\gamma(x), \rho(\gamma)v) \end{aligned}$$

In this definition, we use proposition 2.1. It's easy to see that vector bundle $\tilde{M} \times_{\rho} \mathbb{C}^r \rightarrow M$ is the associated bundle of π_1 -principal bundle $\tilde{M} \rightarrow M$, since the structure group of principal bundle $\tilde{M}$ is discrete, this gives rise to local constant property of transition functions of $\tilde{M} \times_{\rho} \mathbb{C}^r$, which implies $E = \tilde{M} \times_{\rho} \mathbb{C}^r$ carries a flat structure. Eventually, we construct a flat vector bundle E via representation of fundamental group of base manifold M .

Let's pay attention to the situation of principal bundle. Let G be a Lie group and P a G -principal bundle over M . A **flat structure** in principal bundle P is given by $\{U, s_U\}$ such that transition functions g_{VU} are all constant matrixs, and a connection in P is said to be **flat** if its curvature vanishes identically.

Similarly, we can prove that P admits a flat structure iff P admits a flat connection, that is

Theorem 2.1.3. *For a principal G -bundle P over M , the following two conditions are equivalent:*

- 1) P admits a flat structure.
- 2) P admits a flat connection.

Likewise, we attempt to figure out some relations between specific representation of fundamental group π_1 of M and flat connection on G -principal bundle P . Before stating main theorem, we recall some notations.

Denote by $\mathcal{A}(P)$ the set of connections on P , and $\mathcal{A}_{flat}(P)$ the set of flat connections on P . Any gauge transformation $u \in \mathcal{G}(P)$ satisfies

$$u(p.g) = g^{-1}u(p)g$$

$\forall p \in P, g \in G$. We will prove the following theorem

Theorem 2.1.4. *The gauge equivalence class of flat G -connections over a connected manifold M are in one-to-one correspondence with the conjugacy classes of representations $\pi_1(M) \rightarrow G$.*

Proof. Given a connection $D \in \mathcal{A}(P)$ and a loop γ in M based at $x \in M$, consider parallel transport along γ , we have the holonomy map:

$$\tau_{D,x}(\gamma) : P_x \rightarrow P_x$$

Now fix $p \in P$, we can interpret the holonomy as an element in Lie group G as following:

$$\tau_{D,x}(\gamma)p = p \cdot \tau_{D,x,p}(\gamma)$$

Since the action of G on P is free, corresponding element $\tau_{D,x,p}(\gamma)$ in G is unique. Suppose D is a flat connection, then $\tau_{D,x,p}(\gamma)$ depends only on the homotopy class of curve c , this gives rise to a map from $\pi_1(M, x)$ to G :

$$[\gamma] \mapsto \tau_{D,x,p}(\gamma)$$

The above map defines a group homomorphism since

$$\begin{aligned} \tau_{D,x}(\gamma_1 * \gamma_2) &= \text{hol}_{D,x}(\gamma_1) \circ \tau_{D,x}(\gamma_2) \\ \Rightarrow \tau_{D,x,p}(\gamma_1 * \gamma_2) &= \tau_{D,x,p}(\gamma_1) \cdot \tau_{D,x,p}(\gamma_2) \end{aligned}$$

Parallel transport transforms under gauge transformations:

$$\tau_{Du,x,p}(\gamma) = u(p)^{-1} \tau_{D,x,p}(\gamma) u(p)$$

Hence the holonomy map determines a map

$$\tau_{x,p} : \mathcal{A}_{flat}(P)/\mathcal{G}(P) \rightarrow \text{Hom}(\pi_1(M, x), G)/G$$

Where G acts on $\text{Hom}(\pi_1(M, x), G)$ by conjugation. Different choices of x change $\pi_1(M, x)$ by conjugation, hence $\text{Hom}(\pi_1(M, x), G)/G$ doesn't depend on x . Different choices of p change $\text{Hom}(\pi_1(M, x), G)$ by conjugation, hence $\text{Hom}(\pi_1(M, x), G)/G$ doesn't depend on p either.

Finally, we prove the induced map

$$\tau : \sqcup_{[P]} \mathcal{A}_{flat}(P)/\mathcal{G}(P) \rightarrow \text{Hom}(\pi_1(M, x), G)/G$$

is bijective.

- τ is injective

According to Cartan, a connection is determined by its holonomy, therefore τ is injective.

- τ is surjective

Fix $\alpha \in \text{Hom}(\pi_1(M, x), G)$, consider covering map $\tilde{M} \rightarrow M$ and by proposition 1.2.2, π_1 acts on $\tilde{M} \times G$ in the follow method:

$$\begin{aligned} (\tilde{M} \times G) \times \pi_1 &\rightarrow \tilde{M} \times G \\ (x, G)m &\mapsto (m(x), \alpha(m)g) \end{aligned}$$

Then $(\tilde{M} \times G)/\pi_1$ is a G -principal bundle over M , and the trivial flat connection on $\tilde{M} \times G$ descends to a connection D on $(\tilde{M} \times G)/\pi_1$ which satisfies $\tau_{D,x,p} = \alpha$.

□

2 Topics on Chern Class

Let E be a complex vector bundle, $D = d + A$ a connection, and fix a local frame $\{e_\alpha\}$. The curvature corresponding to A is

$$F_A = dA + A \wedge A.$$

Let's check that F_A transforms as a tensor for a change of frame $e'_\alpha = g^\beta_\alpha e_\beta$ (i.e. $e' = eg$).

Put $De' = e'A'$, then

$$\begin{aligned}
 F_{A'} &= dA' + A' \wedge A' \\
 &= d(g^{-1}dg + g^{-1}Ag) + (g^{-1}dg + g^{-1}Ag) \wedge (g^{-1}dg + g^{-1}Ag) \\
 &= -g^{-1}dgg^{-1}dg - g^{-1}dgg^{-1}Ag + g^{-1}dAg - g^{-1}Adg \\
 &\quad + g^{-1}dg \wedge g^{-1}dg + g^{-1}dg \wedge g^{-1}Ag + g^{-1}Ag \wedge g^{-1}dg + g^{-1}Ag \wedge g^{-1}Ag \\
 &= g^{-1}dAg + g^{-1}A \wedge Ag \\
 &= g^{-1}F_Ag.
 \end{aligned}$$

Recall the definition of covariant exterior derivative

$$\begin{aligned}
 D : \Omega^p(E) &\rightarrow \Omega^{p+1}(E) \\
 \omega s &\mapsto d\omega \wedge s + (-1)^{|\omega|} \omega \wedge Ds
 \end{aligned}$$

Consider dual bundle E^* of E , there exists a induced connection D^* on E^* . Specifically, for $s^* \in \Gamma(M, E^*)$ and $s \in \Gamma(M, E)$,

$$\langle D^* s^*, s \rangle \stackrel{set}{=} d\langle s^*, s \rangle - \langle s^*, Ds \rangle$$

Denote connection matrix and curvature matrix of E^* by A^* and Ω^* respectively, then relations between E and E^* reads

$$A^* = -A^T; \quad \Omega^* = -\Omega^T.$$

Now we derive the induced connection and curvature on bundle $\text{End}(E)$. Recall the tensor connection $D^\otimes$ on tensor bundle $E \otimes E^*$ is defined by

$$D^\otimes \stackrel{set}{=} D \otimes \mathbf{1} + \mathbf{1} \otimes D^*,$$

Then $D^\otimes$ induces connection D^{End} on $\text{End}(E)$ via the canonical identification

$$\text{End}(E) \cong E \otimes E^*$$

$$T \mapsto Te_i \otimes e_i^*$$

More specifically,

$$\begin{aligned}
 D^\otimes(Te_i \otimes e_i^*) &= D(Te_i) \otimes e_i^* + Te_j \otimes D^*(e_j^*) \\
 &= D(Te_i) \otimes e_i^* - T(\langle e_j^*, De_i \rangle e_j) \otimes e_i^* \\
 &= D(Te_i) \otimes e_i^* - T(De_i) \otimes e_i^* \\
 &= ((D \circ T - T \circ D)e_i) \otimes e_i^*,
 \end{aligned}$$

Then the canonical identification $\text{End}(E) \cong E \otimes E^*$ reads

$$D^{\text{End}} = D \circ T - T \circ D.$$

Similarly, we can extend connecton $D^{\text{End}} : \Omega^0(\text{End}(E)) \rightarrow \Omega^1(\text{End}(E))$ to $\Omega^*(\text{End}(E))$.

Notice that

$$\begin{aligned}
 D^\otimes(\alpha Te_i \otimes e_i^*) &= D(\alpha Te_i) \otimes e_i^* + (-1)^{|\alpha|} \alpha Te_j \otimes D^*(e_j^*) \\
 &= (d\alpha Te_i + (-1)^{|\alpha|} \alpha \wedge D(Te_i)) \otimes e_i^* - (-1)^{|\alpha|} \alpha T(\langle e_j^*, De_i \rangle e_j) \otimes e_i^* \\
 &= (d\alpha Te_i + (-1)^{|\alpha|} \alpha \wedge D(Te_i)) \otimes e_i^* - (-1)^{|\alpha|} \alpha T(De_i) \otimes e_i^*,
 \end{aligned}$$

which means

$$D^{\text{End}}(\alpha T) = d\alpha T + (-1)^{|\alpha|} \alpha (D \circ T - T \circ D).$$

Now suppose $\omega P \in \Omega^k(\text{End } E)$ and $\eta Q \in \Omega^\ell(\text{End } E)$, we have

$$(-1)^{|\eta|} \omega \wedge \eta(D \circ P) \circ Q = \omega \wedge \eta(D \circ P) \circ Q - \omega \wedge \eta(P \circ D) \circ Q$$

$$\begin{aligned}
 \text{LHS} &= D^{\text{End}}(\omega P \wedge \eta Q) \\
 &= D^{\text{End}}(\omega \wedge \eta P \circ Q) \\
 &= d(\omega \wedge \eta) P \circ Q + (-1)^{|\omega|+|\eta|} \omega \wedge \eta D \circ (P \circ Q) - (-1)^{|\omega|+|\eta|} \omega \wedge \eta(P \circ Q) \circ D \\
 &= d(\omega \wedge \eta) P \circ Q + (-1)^{|\omega|+|\eta|} \omega \wedge \eta(D \circ P) \circ Q \\
 &\quad + (-1)^{|\omega|+|\eta|} \omega \wedge \eta P \circ (D \circ Q) - (-1)^{|\omega|+|\eta|} \omega \wedge \eta(P \circ Q) \circ D
 \end{aligned}$$

$$\begin{aligned}
 \text{RHS} &= (d\omega P + (-1)^{|\omega|}\omega D \circ P - (-1)^{|\omega|}\omega P \circ D) \wedge (\eta Q) \\
 &\quad + (-1)^{|\omega|}(\omega P) \wedge (d\eta Q + (-1)^{|\eta|}\eta D \circ Q - (-1)^{|\eta|}\eta Q \circ D) \\
 &= d\omega \wedge \eta P \circ Q + (-1)^{|\omega|}\omega \wedge \eta(D \circ P) \circ Q - (-1)^{|\omega|}\omega \wedge \eta(P \circ D) \circ Q \\
 &\quad + (-1)^{|\omega|}\omega \wedge d\eta P \circ Q + (-1)^{|\omega|+|\eta|}\omega \wedge \eta P \circ (D \circ Q) \\
 &\quad - (-1)^{|\omega|+|\eta|}\omega \wedge \eta P \circ (Q \circ D) \\
 &= d(\omega \wedge \eta)P \circ Q +
 \end{aligned}$$

Proposition 2.2.1. *The operator D_A on $\text{End}E$ obeys the following identities:*

1. $D_A(\omega \wedge \eta) = D_A\omega \wedge \eta + (-1)^k\omega \wedge D_A\eta$, where $\omega \in \Omega^k(\text{End}E)$ and $\eta \in \Omega^\ell(\text{End}E)$.
2. $d\text{Tr}\omega \wedge \eta = \text{Tr}(D_A\omega \wedge \eta + (-1)^k\omega \wedge D_A\eta)$
3. $d\text{Tr}\omega \wedge \eta \wedge \mu = \text{Tr}(D_A\omega \wedge \eta \wedge \mu + (-1)^k\omega \wedge D_A\eta \wedge \mu + (-1)^{k+\ell}\omega \wedge \eta \wedge D_A\mu)$, where ω , η , and μ are k -, ℓ -, and m -forms, respectively. Similar identities hold for general wedge products.
4. $D_A^2\omega = F_A \wedge \omega + (-1)^{k+1}\omega \wedge F_A$.

Theorem 2.2.2 (Chern-Weil). *For each $p \in \mathbb{N}$, the $2p$ -form*

$$\text{Tr}_K \left(\overbrace{F_A \wedge \cdots \wedge F_A}^{p \text{ times}} \right)$$

is closed, and its cohomology class is independent of the connection A on E .

Proof. Applying the Leibniz rule (3) of Proposition 14.1, we have

$$d\text{Tr}(F_A \wedge \cdots \wedge F_A) = \text{Tr}(D_AF_A \wedge \cdots \wedge F_A) + \text{Tr}(F_A \wedge D_AF_A \wedge \cdots \wedge F_A) + \cdots$$

By Theorem 14.2, each term vanishes.

Next, let A and B be connections on E , and put

$$a = B - A \in \Omega^1(\text{End}E).$$

Let $A_t := A + ta$. Then

$$\frac{d}{dt} \text{Tr}(F_{A_t} \wedge \cdots \wedge F_{A_t}) = \text{Tr}(\dot{F}_{A_t} \wedge F_{A_t} \wedge \cdots \wedge F_{A_t} + F_{A_t} \wedge \dot{F}_{A_t} \wedge \cdots \wedge F_{A_t}) + \cdots$$

$$\begin{aligned}
 &= \operatorname{Tr}(D_A a \wedge F_{A_t} \wedge \cdots \wedge F_{A_t} + F_{A_t} \wedge D_A a \wedge \cdots \wedge F_{A_t} + \cdots) \\
 &= p \operatorname{Tr}(D_A a \wedge F_{A_t} \wedge \cdots \wedge F_{A_t}) \\
 &= p d \operatorname{Tr}(a \wedge F_{A_t} \wedge \cdots \wedge F_{A_t}),
 \end{aligned}$$

where we have used the cyclic property of the trace, the Leibniz rule (3), and the Bianchi identity. Since $A_0 = A$ and $A_1 = B$, we obtain

$$\begin{aligned}
 \operatorname{Tr} F_B \wedge \cdots \wedge F_B - \operatorname{Tr}(F_A \wedge \cdots \wedge F_A) &= \int_0^1 d(p \operatorname{Tr}(a \wedge F_{A_t} \wedge \cdots \wedge F_{A_t})) dt \\
 &= d \left[\int_0^1 p \operatorname{Tr}(a \wedge F_{A_t} \wedge \cdots \wedge F_{A_t}) dt \right].
 \end{aligned}$$

This shows that

$$[\operatorname{Tr} F_A \wedge \cdots \wedge F_A] = [\operatorname{Tr}(F_B \wedge \cdots \wedge F_B)] \in H^{2p}(M),$$

as claimed. $\square$

Corollary 2.2.3. $[\operatorname{Tr}(F_A \wedge \cdots \wedge F_A)] \in H^{2p}(M)$ defines a characteristic class.

Proof. By the previous Theorem, this expression is a well-defined function of the bundle E up to isomorphism. We need only check that it behaves properly under pullback. This is true because $f^* F_A = F_{f^* A}$, which is clear from (13.3), so

$$f^* \operatorname{Tr}(F_A \wedge \cdots \wedge F_A) = \operatorname{Tr}(F_{f^* A} \wedge \cdots \wedge F_{f^* A}).$$

$\square$

Definition 2.2.4. Let E be a vector bundle over $\mathbb{C}$. The first Chern class is given by

$$c_1(E) = \left[\frac{i}{2\pi} \operatorname{Tr}_{\mathbb{C}} F_A \right] \in H^2(M).$$

This agrees with Definition 7.7/8.5 of c_1 .

Proof. It suffices to check that $c_1(\mathcal{O}(-1) \rightarrow \mathbb{CP}^1) = -x$ (exercise: think about why this is sufficient). Here, x is the generator of $H^2(\mathbb{CP}^1, \mathbb{Z}) \cong \mathbb{Z}$ that pairs to one with the fundamental class.

Let $U_0 = (\{z\}, \tau_0)$ and $U_1 = (\{w\}, \tau_1)$, with the frames τ_0, τ_1 given in Example 6.1.3. above. We have $z = \frac{1}{w}$ on the overlap. We showed that $\tau_0 = z\tau_1$, so the transition function is just

$$\sigma = \sigma_{01}(z) = z.$$

Let

$$A^{\tau_0} = \frac{dz\bar{z}}{1 + |z|^2}$$

on U_0 , and

$$A^{\tau_1} = -\frac{dw\bar{w}}{1 + |w|^2}$$

on U_1 .

Claim. This gives a well-defined connection on $\mathcal{O}(-1) \rightarrow \mathbb{CP}^1$.

Proof of claim.

$$\begin{aligned} \sigma A^{\tau_0} \sigma^{-1} - d\sigma \sigma^{-1} &= z A^{\tau_0} z^{-1} - dz z^{-1} \\ &= A^{\tau_0} - dz z^{-1} \\ &= dz\bar{z} \left(\frac{1}{1 + |z|^2} - \frac{1}{|z|^2} \right) \\ &= \frac{dz\bar{z}(|z|^2 - (1 + |z|^2))}{|z|^2(1 + |z|^2)} \\ &= -\frac{dz\bar{z}}{|z|^2(1 + |z|^2)} \\ &= -\frac{1}{w^2} \frac{dw\bar{w}}{\bar{w}(1 + 1/|w|^2)} \\ &= -\frac{dw\bar{w}}{1 + |w|^2} = A^{\tau_1}. \end{aligned}$$

Now, we compute the curvature in the U_0 chart:

$$\begin{aligned} F_A|_{U_0} &= d(A^{\tau_0}) + \underbrace{A^{\tau_0} \wedge A^{\tau_0}}_0 \\ &= \frac{-dz \wedge d\bar{z}}{1 + |z|^2} + \frac{dz\bar{z} \wedge (dz\bar{z} + z d\bar{z})}{(1 + |z|^2)^2} \\ &= \frac{dz \wedge d\bar{z}}{(1 + |z|^2)^2} (-(1 + |z|^2) + |z|^2) \\ &= -\frac{dz \wedge d\bar{z}}{(1 + |z|^2)^2}. \end{aligned}$$

Here, $A^{\tau_0} \wedge A^{\tau_0} = 0$ since this is a $\mathbb{C}$ -valued 1-form.

Let $\mathfrak{g}_E = \Omega^0(\text{End } E)$. Given $u \in \Gamma(\mathfrak{g}_E)$, let $\sigma_t = \exp(tu) \in \Gamma(\mathcal{G}_E)$. Show that

$$\left. \frac{d}{dt}(\sigma_{-t}(A)) \right|_{t=0} = D_A u.$$

Here, $\sigma_t(A)$ is the transformed connection by the usual rule (12.1).

Let A, B be connections and put $a = B - A$. Prove the following identity:

$$\text{Tr}(F_B \wedge F_B - F_A \wedge F_A) = d \text{Tr} \left(D_A a \wedge a + \frac{2}{3} a \wedge a \wedge a \right).$$

The form inside the d on the RHS is called the (relative) Chern-Simons form.

Theorem 2.2.5.

Proposition 2.2.6. *To every complex vector bundle E over a smooth manifold M one can associate a cohomology class $c_1(E) \in H^2(M, \mathbb{Z})$ called **the first Chern class** of E satisfying the following axioms:*

- **(Naturality)** *For every smooth map $f : M \rightarrow N$ and complex vector bundle $E \rightarrow N$, one has $f^*(c_1(E)) = c_1(f^*E)$, where the left is pull-back of cohomology, and f^*E is the pull-back bundle defined by $f^*E_x = E_{f(x)}$.*
- **(Whitney Sum Formula)** *For every bundles E, F over M , one has $c_1(E \oplus F) = c_1(E) + c_1(F)$, where $E \oplus F$ is the Whitney Sum defined as the pull-back of bundle $E \times F \rightarrow M \times M$ via the diagonal inclusion of M in $M \times M$.*
- **(Normalization)** *The first Chern class of the tautological bundle of $\mathbb{CP}^1$ is equal to -1 in $H^2(\mathbb{CP}^1, \mathbb{Z}) \cong \mathbb{Z}$, which means that the integral over $\mathbb{CP}^1$ of any representative of this class equals -1 .*

Theorem 2.2.7. *The k -th Chern class $c_k(E)$ of a complex vector bundle E is represented by the closed $2k$ -form γ_k defined by the following:*

$$\det \left(I_r - \frac{1}{2\pi i} \Omega \right) = 1 + \gamma_1 + \gamma_2 + \cdots + \gamma_r. \quad (2.3)$$

Theorem 2.2.8. *Given any closed real $(1, 1)$ -form φ representing $c_1(E)$, there is a Hermitian structure h in E such that $\varphi = c_1(E, h)$ provided that M is compact Kähler. (In fact, given any h' , a suitable conformal change of h' yields a desired Hermitian structure.)*

Theorem 2.2.9. *Let E be a complex vector bundle of rank r over a manifold M .*

- (a) *If E is flat, then all its Chern classes $c_i(E) \in H^{2i}(M; \mathbb{R})$ are zero for $i \geq 1$;*
- (b) *If E is projectively flat (i.e., satisfies one of the three equivalent conditions in (1.2.7)), then its total Chern class $c(E)$ can be expressed in terms of the first Chern class $c_1(E)$ as follows:*

$$c(E) = \left(1 + \frac{c_1(E)}{r} \right)^r.$$

3 Some Vanishing Results in Complex Geometry

Theorem 2.3.1. *If $\{e_j\}$ is a local orthonormal basis in TM (we identify $\{e_j\}$ with a local basis of $\Lambda^1 M$ via the metric g), and ∇ denotes the Chern connection of E , as well as its prolongation to $\Lambda^{p,q} M \otimes E$ using the Levi-Civita connection on the left hand side of this tensor product. Then operators $\bar{\partial}, \bar{\partial}^*, \nabla^*, \nabla^* \nabla$ can be computed locally by:*

$$\begin{aligned} \bar{\partial} : \Gamma(\Lambda^{p,q} M \otimes E) &\rightarrow \Gamma(\Lambda^{p,q+1} M \otimes E) \\ \sigma &\mapsto \frac{1}{2} (e_j - \sqrt{-1} J e_j) \wedge \nabla_{e_j} \sigma \end{aligned} \quad (2.4)$$

$$\begin{aligned} \bar{\partial}^* : \Gamma(\Lambda^{p,q} M \otimes E) &\rightarrow \Gamma(\Lambda^{p,q-1} M \otimes E) \\ \sigma &\mapsto -\frac{1}{2} (e_j + \sqrt{-1} J e_j) \lrcorner \nabla_{e_j} \sigma \end{aligned} \quad (2.5)$$

$$\begin{aligned} \nabla^* : \Gamma(\Lambda_{\mathbb{C}}^1 \otimes \Lambda^{p,q} M \otimes E) &\rightarrow \Gamma(\Lambda^{p,q} M \otimes E) \\ \omega \otimes \sigma &\mapsto (\delta \omega) \sigma - \nabla_{\omega} \sigma \end{aligned} \quad (2.6)$$

$$\begin{aligned} \nabla^* \nabla : \Gamma(\Lambda^{p,q} M \otimes E) &\rightarrow \Gamma(\Lambda^{p,q} M \otimes E) \\ \sigma &\mapsto \nabla_{\nabla_{e_j} e_j} \sigma - \nabla_{e_j} \nabla_{e_j} \sigma \end{aligned} \quad (2.7)$$

Proof. If (E, h^E) and (F, h^F) are Hermitian bundles, their tensor product inherits a natural Hermitian structure given by

$$h^{E \otimes F}(\sigma^E \otimes \sigma^F, s^E \otimes s^F) := h^E(\sigma^E, s^E) h^F(\sigma^F, s^F).$$

The Hermitian structure on $\Lambda^{p,q}M \otimes E$ with respect to which one defines the adjoint operators above is obtained in this way from the Hermitian structure h of E and the Hermitian structure H of $\Lambda^{p,q}M$ given by (31). We will use the same symbol H for this Hermitian structure, by a slight abuse of notation.

The relation (57) is more or less tautological, using the definition of $\bar{\partial}$ and the fact that $e_j - \sqrt{-1}Je_j$ is a $(1,0)$ -vector, identified via the metric g with a $(0,1)$ -form. Of course, the wedge product there only concerns the $\Lambda^{p,q}M$ -part of σ .

For $\sigma \in C^\infty(\Lambda^{p,q}M \otimes E)$ and $s \in C^\infty(\Lambda^{p,q-1}M \otimes E)$, we define the 1-form α by

$$\alpha(X) := \frac{1}{2}H((X + iJX) \lrcorner \sigma, s).$$

By choosing the local basis $\{e_j\}$ parallel at a point for simplicity, then we get at that point:

$$\begin{aligned} -\delta\alpha &= e_j \lrcorner (\alpha(e_j)) = \frac{1}{2}H((e_j + \sqrt{-1}Je_j) \lrcorner \nabla_{e_j}\sigma, s) + \frac{1}{2}H((e_j + \sqrt{-1}Je_j) \lrcorner \sigma, \nabla_{e_j}s) \\ &= \frac{1}{2}H((e_j + \sqrt{-1}Je_j) \lrcorner \nabla_{e_j}\sigma, s) + \frac{1}{2}H(\sigma, (e_j - \sqrt{-1}Je_j) \wedge \nabla_{e_j}s) \\ &= \frac{1}{2}H((e_j + \sqrt{-1}Je_j) \lrcorner \nabla_{e_j}\sigma, s) + H(\sigma, \bar{\partial}s), \end{aligned}$$

thus showing that the operator $-\frac{1}{2}(e_j + \sqrt{-1}Je_j) \lrcorner \nabla_{e_j}$ is the formal adjoint of $\bar{\partial}$.

The proof of (59) is similar: for $\omega \in C^\infty(\Lambda^\ell M \otimes \Lambda^{p,q}M \otimes E)$ and $s \in C^\infty(\Lambda^{p,q}M \otimes E)$ we define the 1-form α by

$$\alpha(X) := H((\omega(X)) \lrcorner \sigma, s)$$

and compute

$$\begin{aligned} -\delta\alpha &= e_j \lrcorner (\alpha(e_j)) = -H(\delta\omega, s) + H((\omega(e_j)) \lrcorner \nabla_{e_j}s) + H((\omega(e_j)), \nabla_{e_j}s) \\ &= H(\nabla_{e_j}\omega - (\delta\omega), s) + H(\omega, \nabla_{e_j}s), \end{aligned}$$

whence $\nabla^*(\omega \otimes \sigma) = \nabla\sigma$.

Finally, we apply (59) to some section $\nabla = e_j \otimes \nabla_{e_j}$ of $\Lambda^M \otimes \Lambda^M \otimes E$ and get

$$\begin{aligned} \nabla^*\nabla &= (e_j)\nabla_{e_j}\nabla_{e_j} - g(e_j, \nabla_{e_j})\nabla_{e_j} \\ &= g(\nabla_{e_j}e_j, \nabla_{e_j}e_j)\nabla_{e_j}g(e_j). \end{aligned}$$

□

□

Now we use above to derive **The Weitzenböck Formula**:

Theorem 2.3.2. *Let $(E, h) \rightarrow (M^{2m}, g, J)$ be a holomorphic Hermitian bundle over a Kähler manifold M . For vectors $X, Y \in TM$, let $\tilde{R}(X, Y) \in \text{End}(\Lambda^{p,q}M \otimes E)$ be the curvature operator of the tensor product connection on $\Lambda^{p,q}M \otimes E$ induced by the Levi-Civita connection on $\Lambda^{p,q}M$ and the Chern connection on E . Then the following formula holds*

$$2(\bar{\partial}^* \bar{\partial} + \bar{\partial} \bar{\partial}^*) = \nabla^* \nabla + \mathcal{R}, \quad (2.8)$$

where $\mathcal{R}$ is the section of $\text{End}(\Lambda^{p,q}M \otimes E)$ defined by

$$\mathcal{R}(\sigma) := \frac{i}{2} \tilde{R}(Je_j, e_j) \sigma - \frac{1}{2} (e_j - iJe_j) \wedge (e_k + iJe_k) \lrcorner (\tilde{R}(e_j, e_k) \sigma). \quad (2.9)$$

Proof. The proof is a simple computation in a local orthonormal frame parallel at a point, using Lemma 14.1, (57), (58), and (60):

$$\begin{aligned} 2\bar{\partial}^* \bar{\partial} &= -\frac{1}{2} ((e_k + \sqrt{-1}Je_k) \lrcorner \nabla_{e_k} ((e_j - \sqrt{-1}Je_j) \wedge \nabla_{e_j})) \\ &= -\frac{1}{2} ((e_k + \sqrt{-1}Je_k) \lrcorner ((e_j - \sqrt{-1}Je_j) \wedge \nabla_{e_k} \nabla_{e_j})) \\ &= -(g(e_k, e_j) + \sqrt{-1}g(Je_k, e_j)) \nabla_{e_k} \nabla_{e_j} + \frac{1}{2} (e_j - \sqrt{-1}Je_j) \wedge (e_k + \sqrt{-1}Je_k) \lrcorner \nabla_{e_k} \nabla_{e_j} \\ &= \nabla^* \nabla - ig(Je_k, e_j) \nabla_{e_k} \nabla_{e_j} + \frac{1}{2} ((e_j - iJe_j) \wedge (e_k + \sqrt{-1}Je_k) \lrcorner \nabla_{e_k} \nabla_{e_j}) \\ &\quad + \frac{1}{2} ((e_j - \sqrt{-1}Je_j) \wedge (e_k + \sqrt{-1}Je_k) \lrcorner \tilde{R}(e_k, e_j)) \\ &= \nabla^* \nabla - \frac{i}{2} g(Je_k, e_j) \tilde{R}(e_k, e_j) + \frac{1}{2} ((e_j - \sqrt{-1}Je_j) \wedge \nabla_{e_j} ((e_k + \sqrt{-1}Je_k) \lrcorner \nabla_{e_k})) \\ &\quad + \frac{1}{2} ((e_j - \sqrt{-1}Je_j) \wedge (e_k + \sqrt{-1}Je_k) \lrcorner \tilde{R}(e_k, e_j)) \\ &= \nabla^* \nabla - 2\bar{\partial}^* \bar{\partial} + \mathcal{R}. \end{aligned}$$

□

□

Theorem 2.3.3. *Let $(E, h) \rightarrow (M^{2m}, g, J)$ be a holomorphic Hermitian bundle over a Kähler manifold M ,*

Theorem 2.3.4. *Let M be a compact Kähler manifold. If the Ricci curvature of M is negative definite, then M has no holomorphic vector field.*

Theorem 2.3.5. *Let M be a compact Kähler manifold. If the Ricci curvature of M vanishes, then every holomorphic form is parallel. If the Ricci curvature of M is positive definite, then there exists no holomorphic $(p, 0)$ -forms on M for $p > 0$.*

Theorem 2.3.6. *Theorem 1 (Kodaira vanishing). If $\mathcal{L}$ is an ample line bundle on a smooth projective variety X over a field of characteristic zero, then*

$$H^p(X, \Omega^q \otimes \mathcal{L}) = 0 \quad \text{for } p + q > n.$$

We will deduce this result using analytic methods, over $\mathbb{C}$; in fact, the result is false in characteristic p . Let M be a compact Kähler manifold. If L is a holomorphic line bundle on M , we will describe a notion of "positivity" of the line bundle in terms of the Chern classes. We will then have:

Theorem 2.3.7. *If L is a positive line bundle on a compact Kähler manifold M , then*

$$H^p(M, \Omega^q(L)) = 0 \quad \text{for } p + q > n.$$

Moreover, if E is any holomorphic vector bundle on M , then $H^p(M, E \otimes L^r) = 0$ for $r \gg 0$ (if $p > 0$).

4 Connections in Associated Vector Bundles

Chapter 3 Calabi-Yau Theorem and Complex Monge-Ampère Equation

Proposed by Eugenio Calabi in 1954, the Calabi conjecture addresses the relationship between the geometry and topology of **Kähler manifolds**. It consists of two parts:

- **Existence Conjecture:** Let M be a compact Kähler manifold with a closed $(1, 1)$ -form ρ representing the first Chern class $c_1(M)$. Then there exists a unique Kähler metric ω whose Ricci form satisfies:

$$\text{Ric}(\omega) = \rho$$

- **Ricci-Flat Metric Conjecture:** If $c_1(M) = 0$, then M admits a unique Ricci-flat Kähler metric (i.e., a metric with vanishing Ricci curvature).

The conjecture implies that the geometry of Kähler manifolds is entirely governed by their topology. Yau's 1977 proof, known as the **Calabi-Yau Theorem**, transformed the problem into solving a nonlinear PDE (the complex Monge-Ampère equation) and established foundational techniques in geometric analysis. We now give a concise summary of the proof.

let the original Kähler metric be ω , and the target metric be $\tilde{\omega} = \omega + i\partial\bar{\partial}\phi$. The Ricci-flat condition becomes:

$$\det\left(g_{i\bar{j}} + \frac{\partial^2 \phi}{\partial z^i \partial \bar{z}^j}\right) = \text{constant} \cdot \det(g_{i\bar{j}})$$

This is equivalent to the complex Monge-Ampère equation:

$$(\omega + i\partial\bar{\partial}\phi)^n = e^f \omega^n$$

CHAPTER 3 CALABI-YAU THEOREM AND COMPLEX MONGE-AMPÉRE EQUATION

where f satisfies the compatibility condition $\int_M e^f \omega^n = \int_M \omega^n$.

Yau utilized the Continuity Method in his paper. Specifically, Yau constructed a family of equations parameterized by $t \in [0, 1]$:

$$(\omega + i\partial\bar{\partial}\phi_t)^n = e^{t f + c_t} \omega^n$$

with $\phi_0 = 0$. The proof required showing the solution set $\{t\}$ is both open and closed. Furthermore, openness was verified by the implicit function theorem and closedness was verified by a uniform priori estimate on ϕ_t .

Regarding this a priori estimate, the core difficulty lay in establishing three levels of estimates:

- C^0 -estimate: $\|\phi_t\|_{C^0} \leq C$ via the maximum principle.
- C^1 -estimate: $\|\nabla\phi_t\|_{C^0} \leq C$ using the Kähler condition $d\omega = 0$.
- C^2 -estimate: The critical step: $\|i\partial\bar{\partial}\phi_t\|_{C^0} \leq C$. Yau introduced auxiliary functions (e.g., $\log \text{tr}_\omega \tilde{\omega}$) and applied differential inequalities (now called *Yau's inequality*) combined with Ricci identities.
- Higher-order regularity: Bootstrap arguments and Schauder estimates ensured $\phi \in C^\infty$.

Yau's remarkable paper confirmed the deep link between Ricci curvature and topology in Kähler geometry, introduced **Calabi-Yau manifolds**, central to string theory, laid foundations for mirror symmetry and geometric gauge theory and pioneered the field of geometric analysis. His work contained some fundamental innovations, including Geometric analysis framework for complex manifolds, Early inspiration for Ricci flow techniques, Systematic treatment of nonlinear PDEs on Kähler manifolds, Elliptic regularization methods for non-compact spaces, etc $\dots$

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Chapter 4 Hermitian-Yang-Mills

Connections in Stable Vector Bundles

We first summarize the key aspects of the seminal paper “*On the Existence of Hermitian-Yang-Mills Connections in Stable Vector Bundles*” by K. Uhlenbeck and S. T. Yau, published in Communications on Pure and Applied Mathematics, 1986. The paper establishes a fundamental result in complex differential geometry, linking the algebraic geometric concept of stability for holomorphic vector bundles with the analytic concept of Hermitian-Yang-Mills (HYM) connections. This connection, often referred to as the Donaldson-Uhlenbeck-Yau theorem, generalizes earlier work by Donaldson for algebraic surfaces and provides a crucial tool in the study of vector bundles and moduli spaces over Kähler manifolds.

Before the statement of the central theorem, we introduce some definitions.

Definition 4.0.1 (Hermitian-Yang-Mills Connection). *A Hermitian metric h on a holomorphic vector bundle E over a Kähler manifold (X, g) induces a unique connection D compatible with both the metric and the complex structure (the Chern connection). The curvature F of D is a $(1, 1)$ -form with values in $\text{End}(E)$. The metric h (or connection D) is called Hermitian-Yang-Mills if its curvature F satisfies $g^{j\bar{k}} F_{j\bar{k}} = \mu \text{Id}_E$ for some constant μ . This constant is determined by the topology of E and the Kähler class.*

Definition 4.0.2 (Stability). *A holomorphic vector bundle E over a compact Kähler manifold (X, ω) is stable if for every proper coherent analytic subsheaf $\mathcal{F} \subset E$ (with*

$0 < \text{rank}(\mathcal{F}) < \text{rank}(E)$), the following inequality holds:

$$\mu(\mathcal{F}) := \frac{\deg_\omega(\mathcal{F})}{\text{rank}(\mathcal{F})} < \frac{\deg_\omega(E)}{\text{rank}(E)} =: \mu(E)$$

It is semistable if $\mu(\mathcal{F}) \leq \mu(E)$ holds for all such subsheaves. The degree $\deg_\omega(\mathcal{F})$ is defined using the first Chern class of the determinant line bundle of the double dual $\mathcal{F}^{**}$. The paper often assumes stability implies simplicity (i.e., $\dim H^0(X, \text{End}(E)) = 1$).

The central theorem of the paper is:

Theorem 4.0.3 (Uhlenbeck-Yau, 1986). *Let E be a holomorphic vector bundle over a compact Kähler manifold (X, ω) . If E is stable (with respect to ω), then there exists a unique Hermitian metric h on E such that its Chern connection D satisfies the Hermitian-Yang-Mills (HYM) equation:*

$$\sum_{j,k} g^{j\bar{k}} F_{j\bar{k}} = \mu \text{Id}_E$$

where g is the Kähler metric associated with ω , F is the curvature 2-form of D , $g^{j\bar{k}}$ are the components of the inverse metric tensor, $F_{j\bar{k}}$ are the components of the curvature viewed as endomorphisms, Id_E is the identity endomorphism of E , and μ is a constant given by

$$\mu = \frac{2\pi}{\text{vol}(X)\text{rank}(E)} \int_X c_1(E) \wedge \omega^{n-1} = \frac{2\pi}{\text{rank}(E)} \frac{\deg_\omega(E)}{\text{vol}(X)}$$

(Note: The paper uses slightly different normalizations for μ and the Kähler form).

The proof is analytic, relying on solving a non-linear PDE. The core steps are:

1. **Reformulation as a PDE:** Fix a background Hermitian metric h_0 on E . Any other metric h can be written as $h = h_0 k$ where k is a positive definite, self-adjoint endomorphism of E with respect to h_0 . The HYM equation $g^{j\bar{k}} F_{j\bar{k}}(h) = \mu \text{Id}_E$ is rewritten as a non-linear elliptic equation for k :

$$H(k) := H_0 - \text{Tr}_\omega(\bar{\partial}(k^{-1}\partial^0 k)) = 0$$

where $H_0 = H(h_0)$ is the HYM tensor for h_0 (which can be normalized such that $\text{Tr}(H_0) = 0$), Tr_ω denotes the trace using the Kähler metric, and $\partial^0, \bar{\partial}$ are relative to the connection D_0 of h_0 .

2. **Perturbed Equation:** Instead of solving $H(k) = 0$ directly, the authors introduce a perturbed equation involving a parameter $\epsilon > 0$:

$$L_\epsilon(k) := H(k) + \epsilon \log k = 0$$

where $\log k$ is defined via the spectral theorem for k . The term $\epsilon \log k$ acts as a regularizer. It is shown that $\det k = 1$ if $\text{Tr} H_0 = 0$.

3. **Continuity Method (Existence for $\epsilon > 0$):** The existence of a smooth solution k_ϵ to $L_\epsilon(k) = 0$ for all $\epsilon > 0$ is established using the continuity method.

- They show solutions exist for sufficiently large ϵ .
- They prove the set of $\epsilon > 0$ for which solutions exist is both open (using the Implicit Function Theorem by analyzing the linearization δL_ϵ) and closed (by deriving crucial *a priori* estimates on solutions k_ϵ and their derivatives, ensuring convergence as ϵ approaches a limit value $\epsilon_0 > 0$).

4. **Limit as $\epsilon \rightarrow 0$:** The core of the proof lies in analyzing the behavior of the solutions k_ϵ as $\epsilon \rightarrow 0$.

- **Bounded Case:** If the L^2 norm of $\log k_\epsilon$ (or equivalently, the L^2 norm of $H_\epsilon = -\epsilon \log k_\epsilon$) remains bounded as $\epsilon \rightarrow 0$, then standard elliptic estimates imply that a subsequence of k_ϵ converges smoothly to a solution k_0 of the original HYM equation $H(k_0) = 0$.
- **Blow-up Case:** If $\|\log k_\epsilon\|_{L^2} \rightarrow \infty$ as $\epsilon \rightarrow 0$, the authors show this leads to a contradiction with the stability assumption.

5. **Analysis of Blow-up:** When blow-up occurs, a limiting object is constructed.

- Normalize k_ϵ appropriately (e.g., consider $\tilde{k}_\epsilon = p_\epsilon k_\epsilon$ where p_ϵ scales down the maximal eigenvalue). Show uniform L^2_1 bounds on $\tilde{k}_\epsilon$.
- Extract a weakly convergent subsequence in L^2_1 , $\tilde{k}_{\epsilon_j} \rightharpoonup \tilde{k}_\infty$. This limit $\tilde{k}_\infty$ is degenerate (not invertible).
- Construct a limiting projection π (related to the kernel or image of $\tilde{k}_\infty$) which is shown to be orthogonal ($\pi^2 = \pi = \pi^*$) and satisfy a "weak holomorphicity"

condition $(I - \pi)\bar{\partial}^0\pi = 0$ in the sense of distributions. This π represents a subobject of E .

- **Regularity of π :** A crucial step is to show that π defines a proper analytic subsheaf $\mathcal{F} \subset E$. Two independent methods are presented:
 - (a) **Gauge Theory Estimates (Section 5):** Using bounds on the curvature F_ϵ and the HYM tensor H_ϵ , apply Uhlenbeck's compactness theorem for Yang-Mills connections. This yields convergence of the connections away from a small singular set (real codimension ≥ 4), implying π is smooth off this set and defines a reflexive sheaf $\mathcal{F}$.
 - (b) **Direct Regularity (Sections 6 & 7):** Prove a general theorem stating that a "weakly holomorphic map" (defined via holomorphicity on complex line slices) from a complex manifold into an algebraic manifold (like a Grassmannian, where π takes values) must be meromorphic. This again shows π defines a coherent analytic subsheaf $\mathcal{F}$.
- **Contradiction with Stability:** Using the properties of the blow-up limit and the weak holomorphicity of π , the authors derive the inequality $\mu(\mathcal{F}) \geq \mu(E)$ for the subsheaf $\mathcal{F}$ associated with π . Since the blow-up implies $\mathcal{F}$ is a proper non-trivial subsheaf ($0 < \text{rank}(\mathcal{F}) < \text{rank}(E)$), this inequality contradicts the assumption that E is stable ($\mu(\mathcal{F}) < \mu(E)$).

6. **Conclusion:** The blow-up scenario is impossible if E is stable. Therefore, the norms must remain bounded, ensuring convergence to a HYM metric h_0 . Uniqueness follows from the stability condition (simplicity).

The Uhlenbeck-Yau paper provides a powerful existence theorem for Hermitian-Yang-Mills connections on stable vector bundles over arbitrary compact Kähler manifolds. The proof introduces innovative PDE techniques, notably the perturbation method and the analysis of potential blow-ups, linking them directly to the algebraic notion of stability. The result has profound implications, including deriving Bogomolov-type inequalities for Chern classes and providing canonical metrics on stable bundles, which are fundamental in moduli space constructions.